

Weak pure infiniteness and $\mathcal{O}_\infty$ absorption

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Abstract

We prove that every complex C^* -algebra in which each positive element is properly infinite is strongly purely infinite. This answers the ordinary-to-strong part of Kirchberg–Rørdam’s comparison question. For exact algebras, we also prove that proper infiniteness of one fixed finite amplification of every positive element implies proper infiniteness of each positive element. Consequently, every separable nuclear algebra with this fixed-amplification property absorbs $\mathcal{O}_\infty$, without assumptions of unitality or simplicity.

1 Introduction

Pure infiniteness expresses the ability of positive elements of a C^* -algebra to contain several copies of themselves in Cuntz comparison. For a nonsimple algebra, an individual comparison and the simultaneous comparison of two elements are different demands: the latter must also control a mixed entry. We prove that proper infiniteness of every positive element already provides this compatibility. We then give two ways to remove a fixed finite amplification, one for exact algebras and an independent construction for nuclear algebras. In the separable nuclear case the result yields absorption of the Cuntz algebra $\mathcal{O}_\infty$.

1.1 Infiniteness notions and main results

All C^* -algebras are complex and may be nonunital. Let $\mathcal{K}$ denote the compact operators on an infinite-dimensional separable Hilbert space. For $u, v \in (C \otimes \mathcal{K})_+$, write $u \precsim v$ if some sequence $x_j \in C \otimes \mathcal{K}$ satisfies $x_j^* v x_j \rightarrow u$ in norm, and write $u \sim_{\text{Cu}} v$ if comparison holds in both directions. This is stable Cuntz comparison [7]. Finite matrices over C occupy the standard corners of $C \otimes \mathcal{K}$; comparisons between different sizes use zero padding in the other coordinates. Write $h^{\oplus n}$ for the direct sum of n copies of h . A positive element h is *properly infinite* if $h \oplus h \precsim h$. We include zero as properly infinite. For $h \in C_+$, let

$$\text{Her}(h) = \overline{hCh}, \quad \text{Id}(h) = \text{the closed two-sided ideal of } C \text{ generated by } h.$$

We will use the following two comparison conditions:

$$h \text{ is properly infinite for every } h \in C_+, \tag{P}$$

$$h^{\oplus n} \text{ is properly infinite for every } h \in C_+, \quad \text{for one fixed integer } n \geq 1. \tag{F_n}$$

In (F_n) , the same n is chosen before all positive elements are tested. It therefore applies to every scalar spectral cutoff used later. The premise concerns scalar positives in C , with their indicated amplifications.

Condition (P) is equivalent to pure infiniteness in the nonsimple sense of Kirchberg–Rørdam [11, Theorem 4.16]; the equivalence is also recalled in [21, Section 21]. The existence of an n satisfying (F_n) is *weak pure infiniteness* in the fixed-amplification sense. We use the following formulation of *strong pure infiniteness*, valid for arbitrary C^* -algebras [13, Definition 3.1]: for every $x, y \in C_+$ and $\varepsilon > 0$, there are $s, t \in C$ such that

$$\|s^*x^2s - x^2\| < \varepsilon, \quad \|t^*y^2t - y^2\| < \varepsilon, \quad \|s^*xyt\| < \varepsilon. \quad (1.1)$$

The coefficients in this formulation have no prescribed norm bound.

Kirchberg–Rørdam introduced the weak and strong notions in [12]. Strong pure infiniteness implies ordinary pure infiniteness, and (P) implies (F_n) with $n = 1$. The reverse implications form the comparison question in Kirchberg–Rørdam’s Question 9.5, as recalled in [21, Section 21, Problem LXXII]. In version 2 of that problem collection, the question asks whether the three notions agree in general and in the nuclear setting. Our first theorem proves the implication from ordinary to strong pure infiniteness for arbitrary complex C^* -algebras, in the following more flexible form.

Theorem 1.1. *Let C be a complex C^* -algebra satisfying (P). For all $a, b \in C_+$, $c \in C$, and $\varepsilon > 0$, there are $s, t \in C$ such that*

$$\|s^*as - a\| < \varepsilon, \quad \|t^*bt - b\| < \varepsilon, \quad \|s^*ct\| < \varepsilon. \quad (1.2)$$

Consequently C is strongly purely infinite.

The cross term c is arbitrary and need not be a product of the diagonal inputs. Taking $(a, b, c) = (x^2, y^2, xy)$ gives the stated consequence. Arbitrary-entry diagonalization is a known equivalent form of strong pure infiniteness [13, Remarks 3.2(i)–(ii)]; our proof obtains it directly from (P). It also gives coefficient bounds for positive contraction inputs that are independent of the requested mixed-entry accuracy; see Proposition 5.4.

Recall that a C^* -algebra is exact if its minimal tensor product preserves short exact sequences. Exactness supplies the additional uniformity needed to remove the fixed amplification.

Theorem 1.2. *Let C be an exact complex C^* -algebra. Suppose that there is one integer $n \geq 1$ such that $h^{\oplus n}$ is properly infinite for every $h \in C_+$. Then every positive element of C is properly infinite. Consequently C is strongly purely infinite.*

These theorems give a positive answer to the ordinary-to-strong part of Problem LXXII in general and to the full comparison question for exact algebras, hence also for nuclear algebras. The weak-to-ordinary implication for arbitrary nonexact algebras remains outside these conclusions. The statements include nonseparable and nonunital algebras. Section 6 gives an independent proof of the first conclusion of Theorem 1.2 under nuclearity. Its statement is a restriction of that theorem, while its finite-tuple obstruction and merger of hereditary witnesses provide a different method. The same method extends to exact algebras by the deeper theorem that exactness passes to quotients; see Remark 6.7. The direct exact proof avoids that permanence theorem.

Let $\mathcal{O}_\infty$ be the Cuntz algebra generated by countably many isometries with mutually orthogonal ranges. Kirchberg–Rørdam proved the equivalence between strong pure infiniteness and $\mathcal{O}_\infty$ -absorption for separable nuclear algebras that are stable or have an approximate unit of projections [12]. Kirchberg’s later central-sequence results cover general nonunital separable nuclear algebras [10, Proposition 4.4 and Remark 4.6]; see also the paragraph following [13, Definition 3.1]. At that scope they give the following tensorial consequence.

Corollary 1.3. *Let A be a separable nuclear complex C^* -algebra. If there is one integer $n \geq 1$ such that $h^{\oplus n}$ is properly infinite for every $h \in A_+$, then*

$$A \cong A \otimes_{\min} \mathcal{O}_\infty.$$

For separable nuclear algebras, therefore, weak pure infiniteness, pure infiniteness, strong pure infiniteness, and $\mathcal{O}_\infty$ -absorption are equivalent. The reverse implication from absorption to strong pure infiniteness is due to Kirchberg–Rørdam [12]; see also the discussion after [13, Definition 3.1]. Thus a single finite amplification test on positive elements characterizes this tensorial regularity even when the algebra is nonsimple or nonunital.

For example, the cone $C_0((0, 1]) \otimes \mathcal{O}_\infty$ is separable nuclear and absorbs $\mathcal{O}_\infty$ by the self-absorption of $\mathcal{O}_\infty$. It is nonsimple, nonunital, and has no nonzero projections: the norm of a continuous projection-valued function takes only the values zero and one and must tend to zero at the missing endpoint.

1.2 Earlier results and the role of ideal structure

The comparison question has been settled under several structural conditions. Kirchberg–Rørdam treat simple, real-rank-zero, and approximately divisible cases [12]; the real-rank-zero conclusion is also recalled in [21, Section 21]. Blanchard–Kirchberg prove that ordinary pure infiniteness implies strong pure infiniteness when the primitive ideal space is Hausdorff [4, Theorem 5.8]. Rørdam’s proof of [20, Theorem 5.2] gives the same implication for $\mathcal{Z}$ -absorbing purely infinite algebras, where $\mathcal{Z}$ is the Jiang–Su algebra. These two implications do not require exactness or nuclearity.

Pasnicu–Rørdam establish equivalence of the three infiniteness notions for algebras with the ideal property, meaning that projections separate ideals [16, Proposition 2.14]. For algebras of topological dimension zero, Pasnicu–Rouzbehani prove the implication from ordinary to strong pure infiniteness [17, Theorems 3.10(i) and 3.13]. Here topological dimension zero means that the primitive ideal space has a basis of compact open sets. Theorem 1.1 requires no restriction on this space or on the ideal structure.

The weak-to-ordinary implication is closely related to the problem of finding sufficiently many orthogonal equivalent elements in a hereditary subalgebra. Kirchberg–Rørdam showed that the Global Glimm Halving property permits this implication [12, Proposition 4.15]; see [21, Theorem 39]. An equivalent formulation of the Global Glimm Property is that, for every $a \in C_+$ and $\varepsilon > 0$, there is $x \in \text{Her}(a)$ with $x^2 = 0$ such that $(a - \varepsilon)_+$ belongs to the closed two-sided ideal generated by x [22, Remark 3.2 and Theorem 3.6(3)]. Elliott–Rouzbehani prove weak-to-ordinary pure infiniteness in topological dimension zero [8]. Ng–Thiel–Vilalta recover that implication through the Global Glimm Property in a proof that includes nonseparable algebras [15, Theorem 2.3 and Corollary 2.5]. The initial and range squares of a square-zero element are orthogonal and Cuntz equivalent. Our exact argument constructs such elements while retaining a nonzero image throughout a prescribed set of primitive quotients. The relevant sets are the norm slices $\{P \in \text{Prim}(C) : \|h + P\| \geq r\}$, where $h \in C_+$ and $r > 0$; they are quasi-compact even when $\text{Prim}(C)$ is not Hausdorff. Repeated halving on the same slice then supplies enough copies to remove the fixed amplification. The nuclear argument instead combines representation-dependent positive elements into one element that covers a prescribed cutoff.

1.3 Proof strategy and the independent routes

Two tasks drive the argument. To pass from (P) to strong pure infiniteness, one must choose two hereditary corners that retain the relevant parts of their inputs while making the mixed term small.

To remove n from (F_n) , one must first produce sufficiently many equivalent orthogonal pieces inside a scalar hereditary subalgebra. Figure 1 separates these tasks and the two methods for the second one.

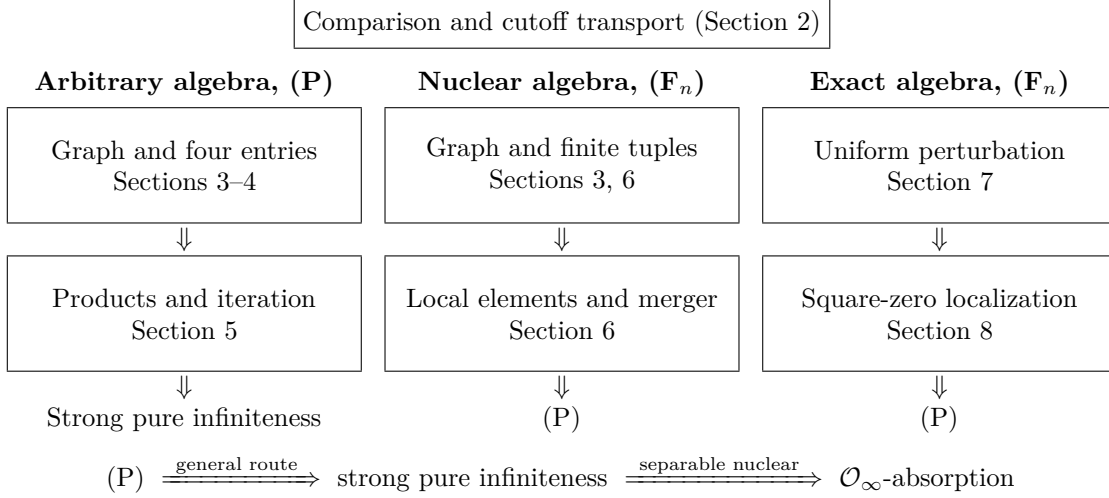


Figure 1: Proof routes and their hypotheses. Comparison transport supports all three routes. Either output (P) can enter the left-hand route. Remark 6.7 extends the middle route to exact algebras using quotient permanence. Under exactness, Section 9 gives an alternative passage from (P) to strong pure infiniteness. The final arrow is the external absorption criterion used in Corollary 1.3.

Comparison transport. Section 2 supplies the common tool. A Cuntz comparison, after a spectral cutoff, can be implemented by a rectangular partial isometry in the bidual. Its products with the cutoff hereditary algebra lie in the original algebra, and conjugation preserves norms in every representation, also at matrix levels. Thus one can place several orthogonal copies in a hereditary subalgebra and transport an entire finite tuple through the same map. These are stronger implementation data than a comparison of classes alone.

From individual to simultaneous comparison. For $h \geq 0$, let e_h be its support projection in the universal bidual. Starting from positive contractions a, b and a contraction c , the first goal is to find positive contractions $p \in \text{Her}(a)$, $q \in \text{Her}(b)$ with

$$\|e_p c e_q\| < 1,$$

while requiring represented norm one of p or q whenever the corresponding input has represented norm greater than $1/2$. This retention of represented norms is what later gives ideal comparison.

Section 3 builds one corner as a graph inside several orthogonal copies of a cutoff of a . The graph has finitely many freely chosen algebra entries. If the positive defect $b(1 - c^*p^2c)b$ vanishes in a representation that detects the relevant part of b , equality in contraction inequalities forces all of those entries to equal fixed measurements, computed before the entries were chosen. Under (P), Section 4 excludes these equalities with four entries. The first pair forces an isometry with a nonzero defect, using the scaling-element construction of Blackadar–Cuntz [3, 16]; the second pair is forced both into and away from that defect. Spectral cutoffs place the relevant supports in actual operator ranges, and a directed family of finite tests makes the contradiction work without separability. Two ideal cutoffs then give the strict support estimate above.

Section 5 upgrades strict contraction to arbitrarily small compression. Uniformly bounded comparison rows show that (P) passes to bounded products. A product contradiction therefore supplies one contraction factor below one, and iteration preserves both represented norm signals while reducing the mixed compression. Finally, high spectral parts of a, b are compared into the separated corners. The coefficients have bounds determined by the diagonal cutoffs, independent of the requested mixed-entry accuracy. This order of choices completes Theorem 1.1.

A separate nuclear proof of amplification removal. Section 6 reuses the graph construction with a different obstruction. A finite tuple witnessing nonexactness has linear matrix norms that no tuple in an exact algebra can realize at every matrix level. Finite-dimensional compressions nevertheless approximate each fixed norm. Under (F_n) , one rectangular carrier transports each entire compressed tuple into $M_n(C)$. If every choice failed to exclude the graph relations, their limiting norms would occur in a quotient of $M_n(C)$. Nuclearity makes this quotient nuclear, hence exact, giving a contradiction. The matrix norm and fixed-element quotient arguments are related to Pisier’s operator-space localization methods [18, pp. 162–165].

It remains to obtain the positive inputs that the graph needs and combine them. The fixed-amplification hypothesis and Kadison transitivity [9] give, at each irreducible representation detecting a positive u , a positive contraction in $\text{Her}(u)$ whose every spectral cutoff is properly infinite and whose norm in that representation is one. This element may depend on the representation. Repeated graph shrinking and a polar deformation merge finitely many such elements while retaining all their norm signals. A finite ideal argument covers each cutoff of u by one merged element and then compares two copies of that cutoff back to u . This proves (P) under nuclearity, independently of the exact argument. The use of hereditary elements to cover ideals is related to the filling-family criteria of [13, Theorem 1.1 and Definitions 4.2, 4.7]; here the merger supplies the simultaneous norm control.

The exact proof and its quantitative completion. Section 7 starts from the nonexact universal unital C^* -algebra of one contraction. Its finite matrix tests and its embedded copy of $C^*(\mathbb{F}_2)$ were studied by Courtney–Sherman [6], with Wassermann’s nonexact free-group example as a predecessor [23]. Finite-word compression at all matrix levels turns nonexactness into one positive error threshold: no contraction in a unital exact algebra can contain approximately reducing copies of every finite matrix contraction. The same copy must approximately intertwine both the contraction and its adjoint.

Combining this obstruction with (F_n) gives a perturbation that prevents cancellation simultaneously in every primitive quotient of a prescribed norm slice. Its lower norm bound is proportional to the perturbation size, with a constant independent of the slice and of n . Section 8 applies this estimate through three finite coupling steps. For a hereditary subalgebra nonzero on a prescribed quasi-compact subset of $\text{Prim}(C)$, it constructs a square-zero element nonzero at every point of that subset. Repeated halving on one fixed norm slice puts n copies of a positive element below the input in Cuntz comparison; ideal comparison and a cutoff limit prove (P). The general theorem then completes Theorem 1.2.

Section 9 gives an alternative final step under exactness. It constructs positives in two prescribed hereditary subalgebras with arbitrarily small product, while retaining a quotient norm floor fixed before the product error is chosen. Two compatible inverse perturbations preserve the two floors. Clipping the later comparison coefficients bounds their norms, so the small product survives implementation and gives strong pure infiniteness directly. Finally, for a separable nuclear algebra, the central-sequence absorption criterion applied in Section 6 gives Corollary 1.3.

2 Comparison and transport

The diagonalization argument repeatedly places several copies of a positive element inside one hereditary subalgebra. Comparison of Cuntz classes alone does not record enough information for this use: the copies must have a common implementation, their products must remain in the algebra, and their norms must agree in every representation. We establish these facts here, together with the ideal and cutoff rules used to retain norm signals.

Cuntz introduced the comparison relation in [7]. The proper-target ideal comparison in Lemma 2.1(iii) is the standard consequence recorded in [11, Proposition 3.5(ii)]; it is also used with that reference in [14, proof of Proposition 2.38]. The cutoff implementation below is related to Rørdam's comparison lemma [19, Proposition 2.4], in the form recalled in [1, Section 2.1]. We give the rectangular and representation statements needed here with their proofs.

All comparisons below are stable: finite positive matrices are placed in a common stabilization by padding with zeros. For a C^* -algebra D and $h \in D_+$, let $\text{Id}_D(h)$ denote the closed two-sided ideal generated by h , and put

$$\text{Her}_D(h) = \overline{hDh}, \quad e_h = \mathbf{1}_{(0,\infty)}(h) \in D^{**}.$$

We omit the subscript when the ambient algebra is clear. Every support projection is taken in the universal bidual. Rectangular matrices over C^{**} are viewed in the corresponding corners of a matrix algebra over C^{**} . The normal extension of a representation sends e_h to $e_{\pi(h)}$: apply normality to $h(h + 1/j)^{-1} \uparrow e_h$. When using a unit or a support projection in a degenerate representation, we restrict to its essential subspace. This leaves the norm of every represented algebra element unchanged.

Lemma 2.1 (Elementary comparison rules). *Let D be a C^* -algebra.*

(i) *For $h \in D_+$,*

$$\text{Her}_D(h) = D \cap e_h D^{**} e_h.$$

If $0 \leq u \leq Kv$ for $u, v \in D_+$ and $K \geq 0$, then $u \in \text{Her}(v)$ and $u \precsim v$. More generally, every positive element of $\text{Her}(v)$ is Cuntz subequivalent to v .

(ii) *For a rectangular algebra matrix z , one has $z^*z \sim_{\text{Cu}} zz^*$. Cuntz subequivalence is transitive, is preserved by block sums, and is closed under norm limits in its first variable when the second variable is fixed. Every $*$ -homomorphism preserves Cuntz subequivalence and equivalence, after amplification. For integers $m, n \geq 1$, $d \in M_n(D)_+$, and $v \in M_m(D)_+$, stable comparison $d \precsim v$ is equivalent to the existence of $r_j \in M_{m,n}(D)$ with $r_j^* v r_j \rightarrow d$ in norm.*

(iii) *If d, v are positive elements in a common stabilization, $d \in \text{Id}(v)$, and v is properly infinite, then $d \precsim v$.*

Proof. We first prove the bidual description in (i), including its norm closure assertion. One inclusion follows on hDh and then by closure. Conversely, suppose $u \in D_+$ and $e_h u = u$. Take increasing continuous positive contractions f_j with $f_j(0) = 0$ and $f_j(t) \uparrow 1$ for $t > 0$. On the compact state space of the unitization, the continuous functions

$$\varphi \longmapsto \varphi(u(1 - f_j(h))u)$$

decrease pointwise to zero. Indeed, the normal extension of each state evaluates the strong limit $f_j(h) \uparrow e_h$. Dini's theorem makes this convergence uniform. The positive norm formula and $(1 - f_j(h))^2 \leq 1 - f_j(h)$ then give $\|(1 - f_j(h))u\| \rightarrow 0$. Thus $u = \lim_j f_j(h)u f_j(h) \in \text{Her}(h)$. Here

$f_j(h) \in \text{Her}(h)$ follows from continuous functional calculus, by first cutting f_j off near zero. For a general element supported on both sides by e_h , apply the positive case to the positive and negative parts of its real and imaginary parts.

If $0 \leq u \leq Kv$, compression by $1 - e_v$ gives $u^{1/2}(1 - e_v) = 0$. Hence u is supported by e_v and belongs to $\text{Her}(v)$. If $u \in \text{Her}(v)_+$, take a cutoff approximate identity $g_j(v)$ for $\text{Her}(v)$, with every g_j zero near zero. The quotient function below is continuous when extended by zero at zero, and gives elements

$$x_j = \frac{g_j(v)}{\sqrt{v}} u^{1/2} \in D, \quad x_j^* v x_j = u^{1/2} g_j(v)^2 u^{1/2} \longrightarrow u.$$

This proves the comparison assertion.

For (ii), the rectangular coefficient $z(z^*z + 1/j)^{-1/2}$ compares z^*z to zz^* , since its conjugation of zz^* equals

$$(z^*z)^2(z^*z + 1/j)^{-1} \longrightarrow z^*z.$$

Apply the same calculation to z^* for the reverse comparison. The inverses lie in unitizations and are multiplied by algebra matrices. Block-diagonal coefficients prove preservation under block sums. For transitivity, first approximate u by x^*vx , then approximate v by y^*wy within the requested error divided by $\max\{1, \|x\|^2\}$; the coefficient yx compares u to w . A diagonal choice of coefficients proves closure under norm limits in the first variable.

If $\phi : D \rightarrow E$ is a $*$ -homomorphism, apply $\phi \otimes \text{id}_{\mathcal{K}}$ to a comparison sequence; continuity preserves its norm limit. Applying this in both directions preserves equivalence. For finite d, v , let P_n, P_m be the scalar coordinate projections in the multiplier algebra of $D \otimes \mathcal{K}$. If $x_j^* v x_j \rightarrow d$ in the stabilization, then

$$r_j = P_m x_j P_n \in M_{m,n}(D), \quad r_j^* v r_j = P_n x_j^* v x_j P_n \longrightarrow d.$$

Conversely, rectangular coefficients embed in the stabilization and give the same convergence there. The multiplier compressions retain algebra membership without a unit, and the argument includes zero matrices.

For (iii), work in the common stabilization D . The closed span of $Dv^{1/2}D$ is $\text{Id}_D(v)$, so finite sums $\sum_{i=1}^r x_i v^{1/2} y_i$ approximate $d^{1/2}$. For such a sum, define

$$X = [x_1 \ \cdots \ x_r] \in M_{1,r}(D), \quad Y = [y_1 \ \cdots \ y_r]^T \in M_{r,1}(D),$$

and put $V = \text{diag}(v, \dots, v) \in M_r(D)$. Then

$$z = \sum_{i=1}^r x_i v^{1/2} y_i = X V^{1/2} Y.$$

The order and rectangular rules already proved give

$$z^*z \leq \|X\|^2 Y^* V Y, \quad Y^* V Y \sim_{\text{Cu}} V^{1/2} Y Y^* V^{1/2} \leq \|Y\|^2 V.$$

It follows that $z^*z \precsim V$. Proper infiniteness, block addition, and induction give $V \precsim v$ for every finite r . Now $z^*z \rightarrow d$, and the norm-limit rule yields $d \precsim v$. This proof includes $v = 0$. $\square$

In particular, Cuntz equivalence preserves proper infiniteness, and a block sum of properly infinite positives is properly infinite. These statements follow from block addition and transitivity. Also, $u \precsim v$ implies $\text{Id}(u) \subset \text{Id}(v)$, since each comparison approximant belongs to $\text{Id}(v)$.

Lemma 2.2 (Transport after a cutoff). *Let C be a C^* -algebra, let $m, n \geq 1$ be integers, and let $d \in M_n(C)_+$ and $v \in M_m(C)_+$. Suppose $d \preceq v$ in $C \otimes \mathcal{K}$. For $\varepsilon > 0$, put $h = (d - \varepsilon)_+$. There is $w \in M_{m,n}(C^{**})$ such that*

$$w^*w = e_h, \quad ww^* \leq e_v, \quad (2.1)$$

and

$$w \operatorname{Her}_{M_n(C)}(h) \subset M_{m,n}(C), \quad w \operatorname{Her}_{M_n(C)}(h)w^* \subset \operatorname{Her}_{M_m(C)}(v). \quad (2.2)$$

Conjugation by w is an injective $*$ -homomorphism on $\operatorname{Her}(h)$. For $l \in \operatorname{Her}(h)_+$,

$$(w l w^* - \tau)_+ = w(l - \tau)_+ w^*, \quad w(l - \tau)_+ w^* \sim_{\text{Cu}} (l - \tau)_+ \quad (\tau \geq 0), \quad (2.3)$$

and

$$w \operatorname{Her}(l)w^* = \operatorname{Her}(w l w^*). \quad (2.4)$$

Every representation, with matching matrix amplifications, satisfies

$$\|\pi^{(m)}(w k w^*)\| = \|\pi^{(n)}(k)\| \quad (k \in \operatorname{Her}(h)). \quad (2.5)$$

The products in these formulas are algebra elements; w itself is only asserted to lie in the bidual.

Proof. Compress a stabilization comparison coefficient to the first m target and first n source coordinates. These scalar multiplier projections give $s \in M_{m,n}(C)$ with $\|d - s^* v s\| < \varepsilon/2$. Put $z = v^{1/2} s h$. Order in the unitization and functional calculus for d give

$$\frac{\varepsilon}{2} h^2 \leq z^* z = h s^* v s h \leq (\|d\| + \varepsilon/2) h^2. \quad (2.6)$$

For the lower bound, use $h d h \geq \varepsilon h^2$. The two inequalities and Lemma 2.1(i) show $\operatorname{Her}(z^* z) = \operatorname{Her}(h)$. Consequently $\operatorname{Her}(|z|) = \operatorname{Her}(h)$ and $e_{z^* z} = e_h$. Let $z = w|z|$ be the rectangular polar decomposition in the bidual. Then $w^* w = e_h$; the factorization through $v^{1/2}$ gives $ww^* = e_{zz^*} \leq e_v$.

For $k \in \operatorname{Her}(h) = \operatorname{Her}(|z|)$, take cutoff functions g_j for $|z|$, zero near zero, with $g_j(|z|)k \rightarrow k$. Since

$$w g_j(|z|) = z \frac{g_j(|z|)}{|z|} \in M_{m,n}(C),$$

norm closure gives $w k \in M_{m,n}(C)$. If $k \geq 0$, then $w k w^* = (w k^{1/2})(w k^{1/2})^*$ belongs to $M_m(C)$. It is supported under e_v , hence belongs to $\operatorname{Her}(v)$. Linear decomposition handles arbitrary k . The identity $w^* w = e_h$ proves multiplicativity and injectivity. It also proves (2.3): functional calculus vanishing at zero commutes with the homomorphism, and the two squares of $w(l - \tau)_+^{1/2}$ are the asserted positives.

For (2.4), normal functional calculus gives $e_{w l w^*} = w e_l w^*$. Thus the image of $\operatorname{Her}(l)$ is supported under $e_{w l w^*}$, which gives one inclusion by Lemma 2.1(i). A sandwich by the image of l has the form

$$(w l w^*) b (w l w^*) = w((w l)^* b (w l)) w^*.$$

The inner algebra element is supported on both sides by e_l , so lies in $\operatorname{Her}(l)$. Since conjugation is an injective $*$ -homomorphism on $\operatorname{Her}(l)$, its image is norm closed; taking closures gives the reverse inclusion. Finally, the normal extension of π carries (2.1) to the same identities with represented supports. Conjugating back by the represented w^* recovers $\pi^{(n)}(k)$. The two contractive conjugations give both inequalities in (2.5). $\square$

If $d_0 \in M_\ell(C)_+$ and $d = d_0^{\oplus r} \in M_{r\ell}(C)_+$, split $w \in M_{m,r\ell}(C^{**})$ into block columns $w_i \in M_{m,\ell}(C^{**})$. Then

$$w_i^* w_j = \delta_{ij} e_{(d_0 - \varepsilon)_+}.$$

The r copies have orthogonal ranges and the same initial support. The single rectangular map also transports every entry of a matrix over the common hereditary algebra. Returning to the notation of Lemma 2.2, for $j \geq 1$ the same support identities apply to $1_j \otimes w \in M_{jm,jn}(C^{**})$. Thus, for every $K \in M_j(\text{Her}_{M_n(C)}(h))$, entrywise conjugation gives an algebra matrix and preserves its norm in every representation, with the matching jm and jn amplifications. This is the joint norm assertion needed when a finite tuple is transported.

For orthogonal $c_1, \dots, c_r \in C_+$, the row $R = [c_1^{1/2}, \dots, c_r^{1/2}] \in M_{1,r}(C)$ satisfies

$$R^* R = \text{diag}(c_1, \dots, c_r), \quad R R^* = \sum_i c_i.$$

Thus their block sum and their sum in C are Cuntz equivalent. Moreover, $f(\sum_i c_i) = \sum_i f(c_i)$ for every continuous f with $f(0) = 0$, first for polynomials with zero constant term and then by continuous functional calculus. Apply this calculation in $M_m(C)$ to the orthogonal transported copies $\theta_i(y) = w_i y w_i^*$ of $y \in \text{Her}_{M_\ell(C)}((d_0 - \varepsilon)_+)$. The cutoff and comparison assertions in Lemma 2.2 give

$$\left(\sum_{i=1}^r \theta_i(y) - \tau \right)_+ \sim_{\text{Cu}} ((y - \tau)_+)^{\oplus r} \quad (\tau \geq 0). \quad (2.7)$$

In particular this includes the zero cutoff.

Lemma 2.3 (Ideals and spectral cutoffs). *Let D be a C^* -algebra.*

- (i) *If $J = \bigcup_\lambda J_\lambda$ for an increasing directed family of closed ideals and $z \in J_+$, then for every $\varepsilon > 0$, $(z - \varepsilon)_+$ lies in some J_λ . In particular, the ideals $\text{Id}((z - \delta)_+)$, directed as $\delta \downarrow 0$, densely exhaust $\text{Id}(z)$.*
- (ii) *For $u, v \in D_+$, one has $u \in \text{Id}(v)$ if and only if every irreducible representation π with $\pi(v) = 0$ also satisfies $\pi(u) = 0$.*
- (iii) *If f, g are continuous nonnegative functions of a positive element a , vanish at zero, and have the same open positivity set in $\text{sp}(a)$, then $\text{Her}(f(a)) = \text{Her}(g(a))$ and $f(a) \sim_{\text{Cu}} g(a)$.*

Proof. For (i), choose $b \in J_\lambda$ with $\|z - b\| < \varepsilon$. Then $\|z + J_\lambda\| < \varepsilon$, and functional calculus in the quotient gives $(z - \varepsilon)_+ \in J_\lambda$. Since $(z - \delta)_+ \rightarrow z$, the particular exhaustion follows. For (ii), apply separation of points by irreducible representations to the quotient $D/\text{Id}(v)$. For (iii), the two support projections coincide. Lemma 2.1(i) gives equality of the hereditary algebras and mutual comparison. $\square$

Definition 2.4. A positive contraction a is *rich* if $(a - \varepsilon)_+$ is properly infinite for every $\varepsilon \geq 0$.

Zero is rich, and under (P) every positive contraction is rich. A *ramp* is a continuous nondecreasing function with values in $[0, 1]$, zero on a neighborhood of zero and constant at 1 beyond a specified larger threshold. A ramp of a rich contraction is rich. Indeed, for $0 \leq \tau < 1$, the open positivity set of $(f(a) - \tau)_+$ is either empty or the positivity set of $(a - \rho)_+$ for some $\rho \geq 0$. Lemma 2.3(iii) and richness apply; for $\tau \geq 1$ the cutoff is zero. Lemma 2.2 gives the same conclusion for transported ramps.

3 Graph measurements and conditional shrinking

For a C^* -algebra C , write $\tilde{C}$ for its minimal unitization when C is nonunital, and set $\tilde{C} = C$ when C is unital. We use its canonical action on C^{**} in products with support projections. Given positive contractions $a, y \in C$, with a rich, and a contraction $c \in \tilde{C}$, we seek positive contractions $p_1 \in \text{Her}(a)$ and $q_1 \in \text{Her}(y)$ with two properties: each has represented norm one wherever its corresponding input has norm greater than $1/2$, and $\|e_{p_1} c e_{q_1}\| < 1$. The norm signals will later give ideal comparison.

The construction has two parts. A graph of orthogonal copies first turns possible failure into finitely many explicit equalities. Once those equalities have been excluded, ideal cutoffs convert the resulting nonvanishing into one global support estimate. We prove both parts here while leaving the finite exclusion problem to the later arguments. Throughout this section put

$$s = \frac{1}{4}, \quad \alpha_0 = \frac{1}{32}.$$

When the integer $k \geq 1$ has been chosen, fix $\gamma_g, \sigma_g > 0$ with

$$\gamma_g^2 + k\sigma_g^2 < 1. \quad (3.1)$$

For example, $\gamma_g = 1/2$ and $\sigma_g = 1/(2\sqrt{k+1})$ satisfy this inequality. Both the number of coordinates and these coefficients are fixed before the measurements below are computed.

Proposition 3.1 (Fixed measurements of a graph). *Let C be a C^* -algebra, let $a, y \in C_+$ be contractions with a rich, and let $c \in \tilde{C}$ be a contraction. For an integer $k \geq 1$ and fixed positive γ_g, σ_g satisfying (3.1), there are*

$$X \in \text{Her}((a - 1/8)_+)_+, \quad B_1, \dots, B_k \in \text{Her}((a - 1/8)_+)$$

with the following property. For any contractions

$$S_i \in \text{Her}((X - \alpha_0)_+), \quad 1 \leq i \leq k,$$

one can form a rich positive contraction $p \in \text{Her}(a)$ such that, for every representation π of C , $\|\pi(a)\| > 1/2$ implies $\|\pi(p)\| = 1$. If a representation π of C satisfies

$$\pi(y(1 - c^* p^2 c)y) = 0, \quad \|\pi(y)\| > s,$$

then, in that same representation,

$$\|\pi(X)\| > s, \quad \pi(B_i) = \pi(S_i) \quad (1 \leq i \leq k). \quad (3.2)$$

The elements $X, B_1, \dots, B_k$ are fixed before the S_i are chosen.

Proof. Let $e = e_{(a-1/8)_+}$ and $H = \text{Her}((a - 1/8)_+)$. Since a is properly infinite, $a^{\oplus(k+2)} \precsim a$. Apply Lemma 2.2 with cutoff $1/8$, and split the resulting row into slots labelled $0, 1, \dots, k, *$. They satisfy

$$v_i^* v_j = \delta_{ij} e, \quad v_i v_i^* \leq e_a, \quad v_i H \subset C. \quad (3.3)$$

Choose a ramp $t = t(a) \in H$ that is zero on $[0, 1/4]$ and one on $[1/2, 1]$. It is rich, and its represented norm is one whenever the represented norm of a exceeds $1/2$.

We now compute the measurements, before making any choice of S_i . Put $F = cyc^* \in C_+$. The block calculation behind their definition is as follows. Suppose $L \in C^{**}$ satisfies $L^*L = e$ and has slot coordinates $v_0^*L = \gamma_g e$, $v_i^*L = \sigma_g S_i$, and suppose $F = LH_0L^*$ for a positive $H_0 \in eC^{**}e$. Then

$$v_0^*Fv_0 = \gamma_g^2 H_0, \quad v_i^*Fv_0 = \sigma_g \gamma_g S_i H_0.$$

Vanishing of the defect in a representation will force this form there, with $tH_0 = H_0$ and with t acting identically on the prescribed supports of S_i . The first measurement therefore recovers H_0 , and an inverse on its spectral cutoff recovers each S_i from the second. These are conditional identities used to design the measurements; their verification follows after the graph is built.

Choose a continuous nonnegative function h_{inv} on $[0, \infty)$, zero at zero, equal to λ^{-1} for $\lambda \geq \alpha_0$, and bounded by α_0^{-1} . Define

$$X = \gamma_g^{-2} t v_0^* F v_0 t, \quad B_i = (\sigma_g \gamma_g)^{-1} t v_i^* F v_0 t h_{\text{inv}}(X). \quad (3.4)$$

The sandwiches are $(v_i t)^* F (v_0 t)$, hence belong to C . They are supported on both sides by e , so they lie in H ; the same is true after multiplication by $h_{\text{inv}}(X)$. Thus $X \geq 0$ and X, B_i have the asserted algebra membership. The slots, t, c, y determine them without using any S_i .

For the chosen S_i , define in the corner $eC^{**}e$

$$J = \left((1 - \gamma_g^2) e - \sigma_g^2 \sum_{i=1}^k S_i^* S_i \right)^{1/2}.$$

Then define in C^{**}

$$L = \gamma_g v_0 + \sigma_g \sum_{i=1}^k v_i S_i + v_* J, \quad p = LtL^*.$$

Here $S_i \in H$, since $X \in H$. The radicand is at least $(1 - \gamma_g^2 - k\sigma_g^2)e$. Its square root belongs to $H + \mathbb{C}e$ and multiplies H into itself. Hence $Lh \in C$ for every $h \in H$, term by term in (3.3). The slot identities also give

$$L^*L = e, \quad e_a L = L.$$

It follows, by the membership and support argument of Lemma 2.2, that $h \mapsto LhL^*$ is an injective $*$ -homomorphism from H into $\text{Her}(a)$. Indeed, for $h \geq 0$ its value is $(Lh^{1/2})(Lh^{1/2})^* \in C$, supported under e_a ; and $L^*L = e$ gives multiplicativity and recovers h on conjugating back. The same identities give preservation of cutoffs, their Cuntz classes, and represented norms, exactly as in that lemma. Thus p is a rich positive contraction with the stated representation signal. If $e = 0$, all the operators on this corner are zero and these assertions still hold.

It remains to see what vanishing of the defect forces. Work on the essential subspace of a representation in which

$$d = y(1 - c^* p^2 c)y = 0, \quad \|y\| > s,$$

and omit the representation symbols. The normal extensions preserve the preceding support identities. For every vector ξ ,

$$\|y\xi\| = \|pcy\xi\| \leq \|cy\xi\| \leq \|y\xi\|.$$

Equality first on $\text{ran } y$ and then on its closure shows that c is isometric on $e_y \mathcal{H}$. For a positive contraction p , equality $\|p\zeta\| = \|\zeta\|$ implies $p\zeta = \zeta$, by applying $1 - p^2$. Consequently $ce_y \mathcal{H}$ lies in the $p = 1$ subspace, and the positive operator $F = cyc^*$ is supported on that subspace. Since

$p = LtL^*$ and $L^*L = e$, the $p = 1$ subspace lies in the range of L , and its pullback lies in the $t = 1$ subspace. With $H_0 = L^*FL$, we obtain

$$F = LH_0L^*, \quad tH_0 = H_0 = H_0t, \quad \|H_0\| = \|F\| = \|y\| > s.$$

The last equality uses the isometric action of c on $e_y\mathcal{H}$.

Now $v_0^*L = \gamma_g e$, $v_i^*L = \sigma_g S_i$, and $L^*v_0 = \gamma_g e$. Substituting these identities into (3.4) yields

$$X = tH_0t = H_0, \quad B_i = tS_iH_0th_{\text{inv}}(X).$$

Since $tX = X$, t is the identity on $\overline{\text{ran } X} = e_X\mathcal{H}$. Each represented S_i has both supports under $e_{(X-\alpha_0)_+}$. Therefore $tS_i = S_i$ and $S_iXh_{\text{inv}}(X) = S_i$. The last display gives $B_i = S_i$, while $\|X\| = \|H_0\| > s$. This proves (3.2). $\square$

Thus it is enough to choose the S_i in their prescribed spectral corner so that the simultaneous relations (3.2) cannot hold. The graph then supplies the nonvanishing implication assumed in the next lemma. The lemma uses the copied element only through its richness and its representation signal, so it is independent of how the finite equalities were excluded.

Lemma 3.2 (From nonvanishing to shrinking). *Let C be a C^* -algebra, let $a, y \in C_+$ be contractions with a rich, and let $c \in \tilde{C}$ be a contraction. Suppose $p \in \text{Her}(a)$ is a rich positive contraction satisfying*

$$\|\pi(a)\| > 1/2 \implies \|\pi(p)\| = 1 \quad \text{for every representation } \pi \text{ of } C.$$

*Put $d = y(1 - c^*p^2c)y \in C_+$, and assume*

$$\pi(d) = 0 \implies \|\pi(y)\| \leq \frac{1}{4} \quad \text{for every representation } \pi \text{ of } C. \quad (3.5)$$

Then there are positive contractions

$$p_1 \in \text{Her}(a), \quad q_1 \in \text{Her}(y)$$

such that p_1 is rich,

$$\begin{aligned} \|\pi(a)\| > 1/2 &\implies \|\pi(p_1)\| = 1, \\ \|\pi(y)\| > 1/2 &\implies \|\pi(q_1)\| = 1 \end{aligned} \quad \text{for every representation } \pi \text{ of } C, \quad (3.6)$$

and

$$\|e_{p_1}ce_{q_1}\|_{C^{**}} < 1. \quad (3.7)$$

The last inequality is a single norm bound in the universal bidual.

Proof. The contraction assumptions give $0 \leq d \leq y^2 \leq 1$. By (3.5), every irreducible representation vanishing on d vanishes on $(y - s)_+$. The primitive ideal test in Lemma 2.3(ii) therefore gives $(y - s)_+ \in \text{Id}(d)$. Set $s' = 3/8$. Apply cutoff exhaustion in Lemma 2.3(i) to $(y - s)_+$ with cutoff $s' - s$. After decreasing the resulting parameter if necessary, there is $0 < \delta < 1$ with

$$(y - s')_+ \in \text{Id}(h_d), \quad h_d = (d - \delta)_+. \quad (3.8)$$

Define

$$z = yh_d^{1/2} \in C, \quad q = zz^* = yh_dy \in \text{Her}(y).$$

The order $d \leq y^2 \leq 1$ gives

$$\delta h_d \leq h_d^{1/2} d h_d^{1/2} \leq z^* z = h_d^{1/2} y^2 h_d^{1/2} \leq h_d. \quad (3.9)$$

Hence $z^* z$ and h_d have the same hereditary algebra, and $z^* z \sim_{\text{Cu}} z z^* = q$. It follows that $\text{Id}(q) = \text{Id}(h_d)$. A second application of cutoff exhaustion, to $(y - s')_+$ with cutoff $1/2 - s'$, gives $\delta' > 0$ such that

$$(y - 1/2)_+ \in \text{Id}((q - \delta')_+). \quad (3.10)$$

Let q_1 be a ramp of q , zero on $[0, \delta'/2]$ and one on $[\delta', \infty)$. It is a positive contraction in $\text{Her}(q) \subset \text{Her}(y)$. If $\|\pi(y)\| > 1/2$, then $\pi((y - 1/2)_+) \neq 0$. The ideal inclusion (3.10) forces $\pi((q - \delta')_+) \neq 0$, so $\|\pi(q)\| > \delta'$ and $\|\pi(q_1)\| = 1$.

To bound the universal-bidual norm, work in the universal nondegenerate representation. For $\xi \in e_{h_d} \mathcal{H} = \mathbf{1}_{(\delta, \infty)}(d) \mathcal{H}$,

$$\begin{aligned} \|p c y \xi\|^2 &= \|y \xi\|^2 - \langle d \xi, \xi \rangle \\ &\leq \|y \xi\|^2 - \delta \|\xi\|^2 \leq (1 - \delta) \|y \xi\|^2. \end{aligned} \quad (3.11)$$

The last inequality uses that y is a contraction. Apply this estimate to vectors in $\text{ran}(h_d^{1/2})$. The closure of $\text{ran}(y h_d^{1/2})$ is $e_q \mathcal{H}$, so continuity gives

$$\|p c e_q\| \leq \sqrt{1 - \delta}.$$

Choose $\sqrt{1 - \delta} < \tau < 1$, and take p_1 to be a ramp of p , zero on $[0, \tau]$ and one at 1. Functional calculus gives $e_{p_1} \leq \tau^{-2} p^2$, while $e_{q_1} \leq e_q$. Consequently

$$\|e_{p_1} c e_{q_1}\| \leq \tau^{-1} \|p c e_{q_1}\| \leq \frac{\sqrt{1 - \delta}}{\tau} < 1. \quad (3.12)$$

The ramp of the rich p is rich and remains in $\text{Her}(a)$. Spectral mapping at its plateau shows that it has norm one in every representation where p has norm one. This proves the first signal in (3.6); the second was proved above. The proof also covers zero inputs. $\square$

4 Four entries under individual proper infiniteness

Assume in this section that C satisfies (P). We choose four entries that rule out the fixed measurement relations in Proposition 3.1. This will supply the nonvanishing hypothesis of Lemma 3.2 and hence a strict contraction between hereditary supports. A spectral cutoff is essential: it places a support in an actual operator range, where the following positivity argument applies.

Lemma 4.1 (A spectral cutoff in an actual range). *Let x, Q be bounded positive operators on a Hilbert space $\mathcal{H}$, and put $U = x Q^2 x$. For every $\eta > 0$,*

$$e_{(U - \eta)_+} \mathcal{H} \subset \text{ran}(x Q), \quad \ker Q \cap \text{ran}(x Q) = \{0\}. \quad (4.1)$$

Proof. Let f_η be the bounded Borel function equal to λ^{-1} for $\lambda > \eta$ and zero otherwise. Its norm on the spectrum is at most η^{-1} . If $\zeta \in e_{(U - \eta)_+} \mathcal{H}$, then

$$\zeta = U f_\eta(U) \zeta = x Q (Q x f_\eta(U) \zeta),$$

which proves the actual-range inclusion. If $\zeta = xQ\xi$ and $Q\zeta = 0$, then

$$0 = \langle QxQ\xi, \xi \rangle = \|x^{1/2}Q\xi\|^2,$$

so $\zeta = xQ\xi = 0$. The assertion concerns the actual range; no closed-range conclusion is being used. $\square$

Proposition 4.2 (Four entries avoid the fixed measurements). *Suppose C satisfies (P). Let $X \in C_+$ and $B_1, B_2, B_3, B_4 \in C$. With*

$$\alpha_0 = \frac{1}{32} < \alpha = \frac{1}{16} < \beta = \frac{1}{8} < s = \frac{1}{4}, \quad h = \frac{1}{2},$$

there exist contractions $T_1, \dots, T_4$ in $\text{Her}((X - \alpha_0)_+)$ for which no representation π of C satisfies

$$\|\pi(X)\| > s, \quad \pi(B_i) = \pi(T_i) \quad (1 \leq i \leq 4). \quad (4.2)$$

Proof. We argue by contradiction. If every four-entry choice could be matched in a representation, the fixed elements could be passed to a norm ultraproduct. In an irreducible representation of the resulting image of C , the first pair would give an isometry with nonzero defect, while the last pair would rule out that defect.

All formulas described as fixed below depend only on X, B_i and the displayed constants. Define

$$x = (X - \alpha)_+, \quad V = (X - \beta)_+, \quad Q = (h1 - B_1 B_1^*)_+, \quad U = xQ^2 x. \quad (4.3)$$

Here $Q \in \tilde{C}_+$, whereas $x, V, U \in C_+$. Let $\text{Fin}(C)$ denote the set of finite subsets of C , and order

$$\Lambda = \mathbb{N}_{\geq 1} \times \text{Fin}(C),$$

by the usual order in the first coordinate and inclusion in the second. At $\lambda = (m, E)$, put $\eta = 1/m$, so $\eta \rightarrow 0$ along this directed set, and every fixed $w \in C$ eventually belongs to E .

The first two entries. The first pair will force $UwV = 0$ for every fixed $w \in C$ in a limiting image of C . At the index (m, E) , we test only the finitely many $w \in E$. A positive sum lets one norm bound control all of these tests: set

$$A_\eta = (U - \eta)_+, \quad D_0 = A_\eta \left(\sum_{w \in E} wVw^* \right) A_\eta, \quad D = g_\eta(D_0), \quad (4.4)$$

where g_η is a ramp that is zero on $[0, \eta]$ and one on $[2\eta, \infty)$. In fact,

$$D_0 = \sum_{w \in E} (A_\eta wV^{1/2})(A_\eta wV^{1/2})^*.$$

Thus a small norm for D_0 makes each $A_\eta wV^{1/2}$ small. We have

$$D_0 \in \text{Her}(A_\eta) \cap \text{Id}(V), \quad e_D \leq e_{(D_0 - \eta)_+} \leq e_{A_\eta} \leq e_x.$$

Since V is properly infinite, ideal comparison gives $D_0 \oplus D_0 \precsim V$. Apply Lemma 2.2 with cutoff η , split the carrier into two columns, and multiply these columns by $D^{1/2}$. Taking adjoints of the resulting two elements gives contractions $T_1, T_2 \in C$ with

$$T_i T_j^* = \delta_{ij} D, \quad e_V T_i^* = T_i^* \quad (i, j = 1, 2). \quad (4.5)$$

Their left supports lie under e_D and their right supports under e_V . Both are under $e_x \leq e_{(X-\alpha_0)_+}$; they therefore belong to the required hereditary algebra.

Consider a representation satisfying (4.2) for the entries at this index, and omit its symbols. Then $B_1 B_1^* = D$ and $Q = (h1 - D)_+$. The spectral subspace of D for (h, ∞) is killed by Q and lies in $e_D \mathcal{H} \subset e_{(U-\eta)_+} \mathcal{H}$. Lemma 4.1 shows that this subspace is zero. Thus

$$B_1 B_1^* = D \leq h1, \quad \|D_0\| \leq 2\eta \quad \text{in this representation.} \quad (4.6)$$

For the second conclusion, if the norm of D_0 were at least 2η , spectral mapping and the plateau of g_η would give $\|D\| = 1 > h$.

A fixed prospective defect and the last two entries. Choose ramps

$$H = h_0(X), \quad K = k_0(X)$$

so that h_0 vanishes on a neighborhood of $[0, \alpha]$ and equals one on $[\beta, \infty)$, and k_0 vanishes on a neighborhood of $[0, \alpha_0]$ and equals one on $[\alpha, \infty)$. In particular $KH = H$. Set

$$r_i = h^{-1/2} B_i^* H \quad (i = 1, 2), \quad b_0 = (1 - H^2)^{1/2}. \quad (4.7)$$

The reason for these definitions is the following conditional calculation. In the limiting irreducible representation below, the first pair will give

$$r_i^* r_j = \delta_{ij} H^2, \quad H r_i = r_i \quad (i, j = 1, 2), \quad H \neq 0.$$

These relations are not being asserted in C at this stage. They imply $b_0 r_i = 0$, and hence

$$v = r_1 + b_0, \quad v^* v = H^2 + b_0^2 = 1, \quad v^* r_2 = 0.$$

Since $r_2^* r_2 = H^2 \neq 0$, the isometry v has a nonzero defect projection $1 - vv^*$. This is the scaling-element construction of Blackadar–Cuntz [3], in the form recalled in [16, Remark 2.4]; the second entry supplies the additional nonzero orthogonal range.

To choose the last two entries before taking any limit, we encode this prospective defect by one fixed element of C . Compression by K puts the formula in the required hereditary algebra, and clipping makes it a positive contraction even before the preceding relations hold. Let $\chi(t) = \min\{1, \max\{0, t\}\}$ on $\mathbb{R}$, and define

$$P_0 = \chi(K[1 - (r_1 + b_0)(r_1 + b_0)^*]K) \in C. \quad (4.8)$$

The compressed selfadjoint element is in C and supported under e_K ; since $\chi(0) = 0$, continuous functional calculus in $\text{Her}(K)$ gives

$$e_{P_0} \leq e_K \leq e_{(X-\alpha_0)_+}.$$

This membership and support statement uses no relation from a representation.

Define the following additional fixed elements:

$$x' = (P_0 - 1/2)_+, \quad Q' = (h1 - B_3 B_3^*)_+, \quad R = x'(Q')^2 x'. \quad (4.9)$$

At the same index η , property (P) gives

$$((R - \eta)_+)^{\oplus 2} \precsim (R - \eta)_+.$$

Apply Lemma 2.2 to this comparison with cutoff η . If t_R is a ramp of R , zero on $[0, 2\eta]$ and one on $[4\eta, \infty)$, the two carrier columns times t_R give contractions $T_3, T_4 \in C$ such that

$$T_3^* T_4 = 0, \quad T_3^* T_3 = T_4^* T_4 = t_R^2. \quad (4.10)$$

Their left supports lie under $e_{(R-\eta)_+}$ and their right supports under $e_{(R-2\eta)_+}$. Both are under $e_{x'} \leq e_{(X-\alpha_0)_+}$. Transport preserves the represented norms, so

$$\|\pi(R)\| > 4\eta \implies \|\pi(T_3)\| = 1. \quad (4.11)$$

In a representation satisfying (4.2), $B_3 = T_3$. Apply Lemma 4.1 to x', Q', R . The spectral subspace of $B_3 B_3^*$ above h is killed by Q' and lies under $e_{(R-\eta)_+}$, so it is zero. Equation (4.11) then gives

$$B_3 B_3^* \leq h1, \quad \|R\| \leq 4\eta \quad \text{in this representation.} \quad (4.12)$$

Passing only fixed elements to an ultraproduct. Suppose, for a contradiction, that every index in Λ admits a representation satisfying (4.2). Choose one, denoted π_λ , on its nondegenerate essential subspace. There is no size issue in this choice: from any such representation, choose a vector detecting $\|\pi_\lambda(X)\| > s$ and restrict to its reducing cyclic subrepresentation. A set of cyclic representations suffices. Choose an ultrafilter ω containing all tails of Λ , and form the norm ultraproduct

$$\mathcal{B} = \prod_{\omega} B(\mathcal{H}_\lambda) = \frac{\prod_{\lambda} B(\mathcal{H}_\lambda)}{\{(z_\lambda) : \lim_{\omega} \|z_\lambda\| = 0\}}.$$

The norm of a class is $\lim_{\omega} \|z_\lambda\|$. Let $\Phi : C \rightarrow \mathcal{B}$ be $\Phi(z) = [(\pi_\lambda(z))]$, and write $B = \Phi(C)$. The image is a closed C^* -subalgebra. The unital extensions carry the unitized formulas into $C^*(B, 1_B)$. We suppress Φ on fixed elements from now on.

For any fixed $w \in C$, eventually $w \in E$. Positivity of the summands in (4.4) and (4.6) yield

$$\|\pi_\lambda((U - \eta)_+ w V^{1/2})\|^2 \leq 2\eta$$

on that tail. Since $\|U - (U - \eta)_+\| \leq \eta$,

$$\|\pi_\lambda(U w V)\| \leq \sqrt{2\eta} \|V\|^{1/2} + \eta \|w\| \|V\| \longrightarrow 0. \quad (4.13)$$

It follows that $UBV = 0$ in the image algebra. The other exact relations and bounds at each index give

$$\begin{aligned} \|X\| \geq s, \quad B_1 B_1^* \leq h1, \quad B_i B_j^* &= \delta_{ij} B_1 B_1^*, \quad H B_i^* = B_i^* \quad (i, j = 1, 2), \\ U B V &= 0, \quad R = 0, \quad B_3 B_3^* \leq h1, \quad B_3^* B_4 = 0, \quad B_3^* B_3 = B_4^* B_4. \end{aligned} \quad (4.14)$$

For $H B_i^* = B_i^*$, use $e_V T_i^* = T_i^*$ and $H = 1$ on the support of V . Fix also a continuous $h_P : [0, 1] \rightarrow [0, 1]$ with $h_P(0) = 0$ and $h_P = 1$ on $[1/2, 1]$. The left support of T_4 lies under $e_{x'}$, so we obtain

$$h_P(P_0) B_4 = B_4. \quad (4.15)$$

Every element in (4.14) and (4.15) is the image of a fixed algebra element or of its fixed unitized functional calculus. We have not placed the varying T_i, D, D_0, t_R in B , and we do not require proper infiniteness of B or of $\mathcal{B}$. The directed finite sets in (4.13) handle all $w \in C$ without a separability assumption.

A nonzero defect gives the contradiction. Since $\|X\| \geq s > \beta$, the image $V = (X - \beta)_+$ is nonzero. Choose an irreducible representation of B in which $V \neq 0$. Extend this nondegenerate representation

to $C^*(B, 1_B) = B + \mathbb{C}1_B$ by sending 1_B to the identity operator; this remains valid if B has a unit different from 1_B . Thus the unitized expressions Q, b_0, Q' and the defect below have their stated represented meanings. Again suppress the representation symbols. The closed span of $BV\mathcal{H}$ is a nonzero reducing subspace, and hence all of $\mathcal{H}$. The relation $UBV = 0$ therefore gives $U = 0$. Put $D_1 = B_1B_1^*$. Since $D_1 \leq h1$, we have $Q = h1 - D_1$, and

$$0 = U = (Qx)^*(Qx) \implies D_1x = hx.$$

By closure $D_1 = h1$ on $e_x\mathcal{H}$. The support of H lies there. Equations (4.7) and (4.14) give

$$r_i^*r_j = \delta_{ij}H^2, \quad Hr_i = r_i \quad (i, j = 1, 2). \quad (4.16)$$

Also $HV = V \neq 0$, so $H \neq 0$ and $r_2 \neq 0$.

The conditional calculation preceding (4.8) now applies: $v = r_1 + b_0$ is an isometry with $v^*r_2 = 0$. Its defect projection

$$P = 1 - vv^*$$

is nonzero, since $Pr_2 = r_2 \neq 0$.

We check that this projection is the represented fixed element P_0 , a point for which both sides of the cutoff matter. The scalar ramps give $KH = H$. From $Hr_1 = r_1$ follows $Kr_1 = r_1$, and the definition $r_1 = h^{-1/2}B_1^*H$ gives $r_1K = r_1$. Also K commutes with b_0 . Expanding the defect gives

$$P = H^2 - r_1r_1^* - r_1b_0 - b_0r_1^*.$$

Each term is fixed by multiplication by K on both sides. Therefore $KPK = P$, and $\chi(P) = P$ because P is a projection. Formula (4.8) yields

$$P_0 = P. \quad (4.17)$$

Now $x' = (P_0 - 1/2)_+ = P/2$. Put $D_3 = B_3B_3^* \leq h1$; then $Q' = h1 - D_3$. The relation $R = 0$ gives $Q'x' = 0$, so

$$D_3P = hP. \quad (4.18)$$

Equations (4.15) and (4.17) say $B_4 = PB_4$. But $B_3^*B_4 = 0$, so $D_3B_4 = 0$, whereas (4.18) gives $D_3B_4 = hB_4$. Thus $B_4 = 0$, and $B_3^*B_3 = B_4^*B_4$ gives $B_3 = 0$. This contradicts (4.18), since $h > 0$ and $P \neq 0$.

The contradiction shows that some index has no representation satisfying (4.2). The four entries chosen at that index prove the proposition. $\square$

We now apply the avoidance proposition to the graph measurements. This gives the form of shrinking that will be used throughout the next section.

Proposition 4.3 (Shrinking under individual proper infiniteness). *Suppose C satisfies (P). Let $a, y \in C$ be positive contractions and let $c \in \tilde{C}$ be a contraction. There are positive contractions $p_1 \in \text{Her}(a)$ and $q_1 \in \text{Her}(y)$, with p_1 rich, such that for every representation π of C ,*

$$\|\pi(a)\| > 1/2 \implies \|\pi(p_1)\| = 1, \quad \|\pi(y)\| > 1/2 \implies \|\pi(q_1)\| = 1, \quad (4.19)$$

and, in the universal bidual,

$$\|e_{p_1}ce_{q_1}\| < 1. \quad (4.20)$$

Proof. Every positive contraction of C is rich under (P), since each of its scalar cutoffs is properly infinite. In Proposition 3.1, fix $k = 4$ before the measurements are computed, and choose, for example, $\gamma_g = 1/2$ and $\sigma_g = 1/(2\sqrt{5})$. Then $\gamma_g^2 + 4\sigma_g^2 < 1$. The graph proposition gives fixed X and $B_1, \dots, B_4$ for the inputs a, y, c . Apply Proposition 4.2 to these elements and use its contractions T_i as the graph choices S_i . The resulting rich positive contraction $p \in \text{Her}(a)$ has the required norm-one signal for a .

Put $d = y(1 - c^*p^2c)y$. If a representation π had $\pi(d) = 0$ and $\|\pi(y)\| > 1/4$, the graph implication would give $\|\pi(X)\| > 1/4$ and $\pi(B_i) = \pi(T_i)$ for every i . These are exactly the relations excluded by Proposition 4.2. Thus

$$\pi(y(1 - c^*p^2c)y) = 0 \implies \|\pi(y)\| \leq 1/4 \quad (4.21)$$

for every representation. The same p is rich and carries the a -signal, so Lemma 3.2 applies to a, y, c, p and proves the stated conclusions. This argument also covers zero inputs. $\square$

5 Uniform separation and general diagonalization

We now prove Theorem 1.1. The strict support bound in Proposition 4.3 initially depends on the inputs. Products turn it into a universal constant below one; iteration then makes the support compression arbitrarily small. A final cutoff comparison gives coefficients whose norms are bounded independently of that requested smallness.

Lemma 5.1 (Products preserve individual proper infiniteness). *Let $(C_i)_{i \in I}$ be any family of C^* -algebras satisfying (P). Their bounded product $D = \prod_{i \in I} C_i$ satisfies (P). More precisely, for a positive contraction $d = (d_i) \in D$ and $0 < \delta < 1/2$, there is a row $s = (s_i) \in M_{1,2}(D)$ such that*

$$s^*ds = (d - 2\delta)_+ \oplus (d - 2\delta)_+, \quad \|s\| \leq \delta^{-1/2}. \quad (5.1)$$

Proof. For an empty index set, $D = 0$ and the zero row has the asserted properties. Otherwise, for each i , put

$$h_i = (d_i - \delta)_+, \quad k_i = (d_i - 2\delta)_+ \oplus (d_i - 2\delta)_+.$$

The positive h_i is properly infinite. Apply Lemma 2.2 to $h_i \oplus h_i \precsim h_i$ with cutoff δ . If v_i is the resulting rectangular partial isometry, then

$$r_i = v_i k_i^{1/2} \in M_{1,2}(C_i), \quad r_i^* r_i = k_i, \quad e_{h_i} r_i = r_i, \quad \|r_i\| \leq 1.$$

The algebra membership follows from transport regularity; no membership of v_i itself is needed. Define

$$\phi_\delta(t) = \begin{cases} t/\delta^{3/2}, & 0 \leq t \leq \delta, \\ t^{-1/2}, & t \geq \delta. \end{cases}$$

This continuous function vanishes at zero, is bounded by $\delta^{-1/2}$, and satisfies $t^{1/2}\phi_\delta(t) = 1$ for $t > \delta$. Consequently

$$s_i = \phi_\delta(d_i) r_i \in M_{1,2}(C_i), \quad d_i^{1/2} s_i = r_i, \quad \|s_i\| \leq \delta^{-1/2}.$$

The uniform bound places (s_i) in $M_{1,2}(D)$. Coordinate functional calculus gives (5.1). Its right-hand side converges to $d \oplus d$ in norm as $\delta \downarrow 0$, proving proper infiniteness of d . Scaling covers every positive element of D . $\square$

The general product-comparison criterion of Antoine–Perera–Thiel [1, Lemma 5.7] also implies this permanence result: for each cutoff ε , one can take a target cutoff $\delta = \varepsilon/2$ and use proper infiniteness in every coordinate. The proof above records the particular bounded rows that make this inference possible. Coordinatewise comparison without a common cutoff bound would not suffice.

Lemma 5.2 (A uniform support contraction). *There exists $\theta \in (0, 1)$ such that for every C^* -algebra C satisfying (P), every pair of positive contractions $a, y \in C$, and every contraction $c \in C$, there are positive contractions $p \in \text{Her}(a)$, $q \in \text{Her}(y)$ such that, for every representation π of C ,*

$$\|\pi(a)\| > 1/2 \implies \|\pi(p)\| = 1, \quad \|\pi(y)\| > 1/2 \implies \|\pi(q)\| = 1,$$

and

$$\|e_p c e_q\| \leq \theta. \tag{5.2}$$

Proof. If no such bound existed, for each $m \geq 2$ we could choose an algebra C_m satisfying (P) and contractions a_m, y_m, c_m , with the first two positive, such that every pair with the stated membership and simultaneous signals has support norm greater than $1 - 1/m$. Form their bounded product D and the three elements $a = (a_m)$, $y = (y_m)$, $c = (c_m) \in D$. Lemma 5.1 gives (P) for D , so Proposition 4.3 produces p, q with the signals and $\kappa = \|e_p c e_q\| < 1$.

For each m , the coordinate homomorphism $D \rightarrow C_m$ preserves hereditary membership, and its normal extension sends the two support projections to the coordinate supports. Thus the coordinate support norm is at most κ . If π is any representation of C_m , its composition with the coordinate homomorphism is a representation of D , so both signals hold for the coordinate pair as well. Choosing m with $1 - 1/m > \kappa$ contradicts its selection. The resulting constant can be increased, if necessary, to lie in $(0, 1)$. $\square$

This proof uses the normal coordinate maps, not an identification of D^{**} with a product of biduals. The input c in Lemma 5.2 belongs to C . This ensures that the sequence (c_m) belongs to D ; a sequence of arbitrary individual unitizations need not belong to D or its unitization. We need the uniform statement only in the stated form.

Proposition 5.3 (Arbitrarily small support compression). *Suppose C satisfies (P). For positive contractions $a_*, b_* \in C$, a contraction $c_* \in C$, and $\zeta > 0$, there are positive contractions $p \in \text{Her}(a_*)$, $q \in \text{Her}(b_*)$ such that*

$$\|e_p c_* e_q\| < \zeta$$

and, for every representation π ,

$$\|\pi(a_*)\| > 1/2 \implies \|\pi(p)\| = 1, \quad \|\pi(b_*)\| > 1/2 \implies \|\pi(q)\| = 1.$$

Proof. Fix θ from Lemma 5.2. On $[0, 1]$ set

$$f(t) = (2t - 1)_+, \quad g(t) = \min\{1, (4t - 1)_+\}.$$

Both vanish near zero, $f(1) = 1$, and $g = 1$ on $[1/2, 1]$. For a positive contraction h ,

$$g(h) = e_h g(h), \quad g(h) e_{f(h)} = e_{f(h)}. \tag{5.3}$$

Apply Lemma 5.2 once to a_*, b_*, c_* , obtaining p_1, q_1 with support norm at most θ . Suppose inductively that $\|e_{p_k} c_* e_{q_k}\| \leq \theta^k$. The next normalized input is

$$c^{(k)} = \theta^{-k} g(p_k) c_* g(q_k) \in C, \quad \|c^{(k)}\| \leq 1. \tag{5.4}$$

Apply the uniform lemma to $f(p_k), f(q_k), c^{(k)}$. The new supports lie under those of $f(p_k)$ and $f(q_k)$; by (5.3),

$$e_{p_{k+1}} c_* e_{q_{k+1}} = \theta^k e_{p_{k+1}} c^{(k)} e_{q_{k+1}}, \quad \|e_{p_{k+1}} c_* e_{q_{k+1}}\| \leq \theta^{k+1}.$$

The hereditary algebras are nested. In every representation, norm one of p_k implies norm one of $f(p_k)$, by spectral mapping and $f(1) = 1$, and similarly for q_k . Thus the two original signals persist at every step. All positive contractions needed as first inputs are rich under (P). Choose k with $\theta^k < \zeta$. $\square$

Proposition 5.4 (Comparison with bounded coefficients). *Suppose C satisfies (P). Let $a, b \in C_+$ be contractions and let $c \in C$. Given*

$$0 < \delta < 1/2, \quad 0 < \rho \leq 2\delta, \quad \zeta > 0,$$

there exist $s, t \in C$ such that

$$\|s^* a s - a\|, \|t^* b t - b\| < 2\delta + \rho, \quad (5.5)$$

$$\|s^* c t\| < \zeta, \quad \|s\|, \|t\| \leq \delta^{-1/2}. \quad (5.6)$$

Proof. Let a_*, b_* be ramps of a, b , zero on $[0, \delta]$ and one on $[2\delta, 1]$. Put $M_c = \max\{1, \|c\|\}$. Choose a finite iteration in Proposition 5.3 so that its outputs satisfy the signals and

$$\|e_p c e_q\| < \delta \zeta. \quad (5.7)$$

For example, apply the construction to c/M_c and take k with $M_c \theta^k < \delta \zeta$. This choice precedes the comparison coefficients below.

It remains to realize the diagonal cutoffs through p and q with a coefficient bound depending only on δ and ρ .

Put $d_a = (a - 2\delta)_+$. If an irreducible representation detects d_a , its norm on a is greater than 2δ . It then has norm one on a_* , and the signal gives norm one on p . Lemma 2.3(ii) yields $d_a \in \text{Id}(p)$. Since

$$e_p \leq e_{a_*} \leq \mathbf{1}_{(\delta, \infty)}(a), \quad pap \geq \delta p^2,$$

we have $\text{Id}(pap) = \text{Id}(p)$: one inclusion is immediate, and the displayed order gives the other. The positive pap is properly infinite under (P), so Lemma 2.1(iii) gives $d_a \precsim pap$.

Compressing a stabilization comparison coefficient to the scalar source and target corners supplies $w \in C$ with

$$\|w^* pap w - d_a\| < \rho.$$

Set $s = pw$. Then $e_p s = s$, $s^* a s = w^* pap w$, and

$$\delta s^* s \leq s^* a s, \quad \|s\|^2 \leq \frac{\|d_a\| + \rho}{\delta} \leq \frac{1 - 2\delta + \rho}{\delta} \leq \frac{1}{\delta}. \quad (5.8)$$

In particular the more general estimate $\|pw\|^2 \leq (1 + \rho)/\delta$ holds without the restriction $\rho \leq 2\delta$. The same argument for b, q gives $t = qw'$, $e_q t = t$, and the identical bounds. Because $\|a - d_a\| \leq 2\delta$, these comparisons prove (5.5). Finally

$$\|s^* c t\| \leq \|s\| \|t\| \|e_p c e_q\| \leq \delta^{-1} \|e_p c e_q\| < \zeta.$$

The bound in (5.8) depends on δ, ρ , not on p, q or the separation accuracy. This is why the preceding choice (5.7) remains valid after the comparison coefficients are found. $\square$

Proof of Theorem 1.1. For arbitrary $a, b \in C_+$, $c \in C$, and $\varepsilon > 0$, put $M = \max\{1, \|a\|, \|b\|\}$. Choose $0 < \delta < 1/2$ and $0 < \rho \leq 2\delta$ with $M(2\delta + \rho) < \varepsilon$. Apply Proposition 5.4 to $a/M, b/M, c$ with $\zeta = \varepsilon$. Multiplying the two diagonal inequalities by M gives all three inequalities in (1.2).

For positive $x, y \in C$, specialize these inequalities to $(a, b, c) = (x^2, y^2, xy)$. They are precisely (1.1), so C is strongly purely infinite. This uses the unrestricted definition directly. If one instead writes a block matrix $Z = \begin{pmatrix} a & c \\ c^* & b \end{pmatrix}$, the norm of $\text{diag}(s, t)^* Z \text{diag}(s, t) - (a \oplus b)$ is at most the maximum of the two diagonal errors plus $\|s^* c t\|$. Hence the same estimates also give diagonal approximation of every such block matrix, in particular of every positive one. The argument includes zero elements and the zero algebra. $\square$

6 Removing amplification under nuclearity

We now remove the fixed amplification under nuclearity. This proof uses the graph construction of Section 3: a finite tuple of matrix entries rules out the graph's failure relations, and the resulting shrinking operation allows us to join rich positive elements detected by different representations. Its second output need not be rich; the merger retains n orthogonal copies so that (F_n) makes their sum rich. A finite ideal argument then reaches every positive element. The proof of the following theorem is independent of the exact localization argument in Sections 7–8.

Theorem 6.1 (Removing amplification under nuclearity). *Let C be a nuclear complex C^* -algebra. Suppose that one integer $n \geq 1$, fixed for the whole algebra, makes $h^{\oplus n}$ properly infinite for every $h \in C_+$. Then every positive element of C is properly infinite.*

Theorem 6.1 assumes neither separability nor a unit. We keep the fixed integer n throughout this section. In the tuple construction it controls the number of graph entries, while the dimensions of the finite matrix approximations may vary.

6.1 A finite tuple that exact algebras cannot realize

The graph measurements are fixed before their entries are chosen. To choose those entries under nuclearity, we use a finite tuple whose linear matrix norms cannot all occur in an exact algebra, even though each fixed such norm is a limit of norms of finite matrix tuples. Nonexact examples originate in work of Wassermann [23]. Pisier develops the operator-space formulation and finite compression mechanism in [18, Theorems 1 and 7, Lemma 2 and Equation (7)]. We use the existence of a nonexact algebra from this theory and give the tensor norm and compression arguments needed here. They retain the quantifiers across all matrix levels required for the graph entries.

Recall that a C^* -algebra D is *exact* if tensoring every short exact sequence with D in the minimal tensor norm preserves exactness. Thus exactness identifies the quotient norm on $(D \otimes_{\min} J)/(D \otimes_{\min} I)$ with the spatial norm on $D \otimes_{\min} (J/I)$, for every closed ideal $I \triangleleft J$. The next proof starts from a strict inequality between these norms.

Lemma 6.2 (A finite tuple and finite matrix approximations). *There are $\ell \geq 1$, a C^* -algebra W , contractions $z_1, \dots, z_\ell \in W$, and contractions $z_j^{(r)} \in M_{N_r}(\mathbb{C})$, where $N_r \geq 1$, $r \geq 1$, and $1 \leq j \leq \ell$, with the following properties.*

(i) *No tuple $b_1, \dots, b_\ell$ in any exact C^* -algebra satisfies*

$$\left\| \sum_{j=1}^{\ell} A_j \otimes b_j \right\| = \left\| \sum_{j=1}^{\ell} A_j \otimes z_j \right\| \quad \text{for every } m \geq 1 \text{ and every } A_j \in M_m(\mathbb{C}). \quad (6.1)$$

(ii) For every fixed $m \geq 1$ and $A_j \in M_m(\mathbb{C})$,

$$\lim_{r \rightarrow \infty} \left\| \sum_{j=1}^{\ell} A_j \otimes z_j^{(r)} \right\| = \left\| \sum_{j=1}^{\ell} A_j \otimes z_j \right\|. \quad (6.2)$$

Proof. Take a nonexact C^* -algebra W , whose existence is recorded in [18, Remark (i) following Theorem 1, pp. 163–164]. There are $I \triangleleft J$ such that the canonical surjection

$$(W \otimes_{\min} J)/(W \otimes_{\min} I) \longrightarrow W \otimes_{\min} (J/I)$$

is not injective. The map is surjective because its algebraic image is dense and the range of a C^* -homomorphism is closed. Choose a representative $T_0 \in W \otimes_{\min} J$ of a nonzero class in its kernel. Its norm modulo $W \otimes_{\min} I$ is positive, whereas its image in $W \otimes_{\min} (J/I)$ is zero. A sufficiently close finite algebraic approximation to T_0 retains a strict gap between these two norms. After normalizing its nonzero first factors and absorbing their norms into the second factors, we obtain contractions $z_j \in W$ and elements $h_j \in J$ for which

$$\left\| \sum_j z_j \otimes h_j + W \otimes_{\min} I \right\| > \left\| \sum_j z_j \otimes (h_j + I) \right\|_{\min}. \quad (6.3)$$

For a tuple $\mathbf{v} = (v_j)_{j=1}^{\ell}$ in a C^* -algebra D , write

$$\mu_{\mathbf{v},m}(A_1, \dots, A_{\ell}) = \left\| \sum_{j=1}^{\ell} A_j \otimes v_j \right\|_{M_m(D)}.$$

We show that these functions determine both norms in (6.3), with the coefficient algebra and its ideal held fixed.

For arbitrary coefficients $d_j \in J$, represent the fixed nonzero algebra J faithfully on a Hilbert space $\mathcal{H}$. Then

$$\left\| \sum_j v_j \otimes d_j \right\|_{\min} = \sup_{m,V} \mu_{\mathbf{v},m}(V^* d_1 V, \dots, V^* d_{\ell} V), \quad (6.4)$$

where $V : \mathbb{C}^m \rightarrow \mathcal{H}$ ranges over finite-dimensional isometries. Indeed, also represent D faithfully. The operators on the right are the two-sided compressions in the second Hilbert factor, so their norms are at most the left-hand norm. Finite-rank projections, directed by inclusion, converge strongly to the identity. Their two-sided compressions converge strongly to the tensor sum; evaluation at a fixed vector whose image nearly attains its norm gives the reverse inequality. This applies to arbitrary, including nonselfadjoint, tensor sums.

For the quotient norm, choose a positive contractive approximate identity (f_{λ}) of the fixed ideal I . Put $J_D = D \otimes_{\min} I$. For $T \in D \otimes_{\min} J$, define

$$R_{\lambda}(T) = (1 \otimes (1 - f_{\lambda}))T(1 \otimes (1 - f_{\lambda})),$$

using the unitized multiplier operations. Elementary tensors and density give

$$T - R_{\lambda}(T) \in J_D, \quad \|R_{\lambda}(K)\| \longrightarrow 0 \quad \text{for each fixed } K \in J_D.$$

The cutdowns are contractions. Hence, for every fixed $K \in J_D$,

$$\text{dist}(T, J_D) \leq \|R_{\lambda}(T)\| \leq \|T - K\| + \|R_{\lambda}(K)\|.$$

Taking lower and upper limits and then the infimum over K yields the net limit

$$\left\| \sum_j v_j \otimes d_j + J_D \right\| = \lim_{\lambda} \left\| \sum_j v_j \otimes (1 - f_{\lambda}) d_j (1 - f_{\lambda}) \right\|_{\min}. \quad (6.5)$$

The cutdown coefficients lie in J , and no quas centrality is needed. This argument uses convergence on each fixed element of J_D . A related fixed-element cutdown mechanism appears in [18, proof of Lemma 3, p. 165]; the calculation above establishes the stated formula for general $I \triangleleft J$ and its approximate-identity net. Equations (6.4) and (6.5) show that the displayed quotient norm depends only on the functions $\mu_{\mathbf{v},m}$. Applying the same compression argument in J/I shows the same for $\| \sum_j v_j \otimes (d_j + I) \|_{\min}$.

If a tuple in an exact algebra had the same functions $\mu_{\mathbf{z},m}$ at every matrix level, the two formulas above would transfer this strict gap to that algebra. Exactness makes its canonical quotient map isometric, a contradiction. This proves (i). The conclusion excludes simultaneous equality for all finite matrix tests; it does not assert that a fixed finite list of tests suffices.

For (ii), set $W_0 = C^*(z_1, \dots, z_{\ell})$. This algebra is separable, and its inclusion in W preserves matrix norms. It has a faithful representation on a separable Hilbert space. Indeed, choose a dense sequence in the positive unit sphere and, for each member, a state attaining its norm. Their GNS representations have faithful direct sum: for every nonzero positive element, approximation by that sequence shows that some chosen state is positive on it. Choose increasing nonzero finite-rank projections $P_r \rightarrow 1$ strongly in that representation, and set

$$N_r = \dim P_r \mathcal{H}, \quad z_j^{(r)} = P_r z_j P_r|_{P_r \mathcal{H}}.$$

These are contractions. For each fixed m and (A_j) , the corresponding two-sided compressions of $\sum_j A_j \otimes z_j$ converge strongly to that operator and have norms at most its norm. Evaluation at a fixed near-norming vector gives the matching lower limit, proving (6.2). The same sequence works for every fixed test. The assertion is pointwise in the test, and uses no multiplicativity of the compressions. $\square$

6.2 Choosing graph entries under nuclearity

We use the finite tuples from Lemma 6.2 to choose entries in a spectral hereditary subalgebra. The central point is that the transport lemma provides one rectangular map for the whole tuple. Its conjugation preserves the norm of every joint linear matrix expression, not merely the norms of the individual entries.

Proposition 6.3 (Nuclear avoidance of the fixed measurements). *Let C be nuclear and satisfy (F_n) for a fixed $n \geq 1$, and let ℓ be as in Lemma 6.2. For $X \in C_+$ and fixed elements $B_{j,pq} \in C$, where $1 \leq j \leq \ell$ and $1 \leq p, q \leq n$, there are contractions*

$$S_{j,pq} \in \text{Her}((X - \alpha_0)_+)$$

such that no representation π satisfies simultaneously

$$\|\pi(X)\| > s, \quad \pi(B_{j,pq}) = \pi(S_{j,pq}) \quad (1 \leq j \leq \ell, 1 \leq p, q \leq n). \quad (6.6)$$

Here $\alpha_0 = 1/32$ and $s = 1/4$, as in Proposition 3.1.

Proof. Put $g = (X - \alpha_0)_+$, choose $0 < \nu < s - \alpha_0$, and put $h_g = (g - \nu)_+$. Choose a ramp $t' = t'(X)$ that is zero on $[0, \alpha_0 + \nu]$ and one on $[s, \infty)$. Then $t' \in \text{Her}(h_g)$, and

$$\|\pi(X)\| > s \implies \|\pi(t')\| = 1. \quad (6.7)$$

The element $v = g^{\oplus n} \in M_n(C)$ is properly infinite by (F_n) , applied to the scalar positive g . For the dimensions N_r of Lemma 6.2,

$$d_r = g^{\oplus N_r} \precsim v^{\oplus N_r} \precsim v.$$

The first comparison uses g as a summand of v ; the second uses proper infiniteness of v . Lemma 2.2, with cutoff ν , gives one rectangular carrier

$$w_r \in M_{n, N_r}(C^{**}), \quad w_r^* w_r = 1_{N_r} \otimes e_{h_g}, \quad w_r w_r^* \leq 1_n \otimes e_g.$$

Its conjugation sends $M_{N_r}(\text{Her}(h_g))$ into $M_n(\text{Her}(g)) = \text{Her}_{M_n(C)}(v)$. Use this same carrier for every j , defining

$$S_j^{(r)} = w_r(z_j^{(r)} \otimes t')w_r^* \in M_n(\text{Her}(g)), \quad S_{j,pq}^{(r)} = (S_j^{(r)})_{pq}. \quad (6.8)$$

Each matrix has norm at most one, so its entries are contractions in the required hereditary subalgebra.

Suppose that every r admits a representation π_r satisfying (6.6) for these entries. Restricting to a reducing cyclic subrepresentation that still detects $\|\pi_r(X)\| > s$, we may make these choices from a set of representations. Write $B_j = [B_{j,pq}] \in M_n(C)$; these elements are fixed independently of r . At every matrix level m , the normal amplification of the single w_r is isometric on the supported source algebra, by Lemma 2.2. The represented entry equalities and (6.7) therefore give, for every $A_j \in M_m(\mathbb{C})$,

$$\begin{aligned} \left\| \sum_j A_j \otimes \pi_r^{(n)}(B_j) \right\| &= \left\| \left(\sum_j A_j \otimes z_j^{(r)} \right) \otimes \pi_r(t') \right\| \\ &= \left\| \sum_j A_j \otimes z_j^{(r)} \right\|. \end{aligned} \quad (6.9)$$

This equality is exact at each r . Using a different carrier for each j would not give the joint equality.

Define the $*$ -homomorphism

$$\Psi : M_n(C) \longrightarrow \mathcal{Q} = \frac{\prod_r B(\mathcal{H}_r^{\oplus n})}{\bigoplus_r B(\mathcal{H}_r^{\oplus n})}, \quad \Psi(b) = [(\pi_r^{(n)}(b))_r],$$

where the numerator is the algebra of bounded sequences and the denominator is its ideal of norm-null sequences. Contractivity of the representations makes each displayed sequence bounded. The image $E = \Psi(M_n(C))$ is closed and isomorphic to $M_n(C)/\ker \Psi$. Since $M_n(C) \cong M_n \otimes C$, nuclearity of matrix algebras and permanence under tensor products make $M_n(C)$ nuclear [2, II.9.4.2 and IV.3.1.1]. Its quotient E is nuclear by [5, Corollary 4], and hence exact by [2, II.9.6.6]. These permanence facts require neither units nor separability. It is this quotient of a nuclear algebra that must be exact; the argument requires no exactness of the ambient $\mathcal{Q}$.

The norm of a class modulo norm-null sequences is the limsup of its coordinate norms. Subtracting a norm-null sequence gives the lower bound, and subtracting finitely many initial entries gives the upper bound. The same formula holds at each fixed matrix level under the canonical

identifications of finite matrices over a product and its norm-null ideal. Since $E \subset \mathcal{Q}$ is completely isometric, (6.9) and (6.2) give

$$\left\| \sum_j A_j \otimes \Psi(B_j) \right\|_{M_m(E)} = \limsup_r \left\| \sum_j A_j \otimes \pi_r^{(n)}(B_j) \right\| = \left\| \sum_j A_j \otimes z_j \right\|$$

for every m and every (A_j) . Only the fixed elements B_j have been sent into E ; the varying choices in (6.8) were used in the coordinatewise equalities. The displayed identity contradicts Lemma 6.2(i). Thus some r admits no such representation, and its entries in (6.8) prove the proposition. $\square$

Proposition 6.4 (Shrinking two corners under nuclearity). *Let C be nuclear and satisfy (F_n) for one fixed $n \geq 1$. Let $a, y \in C_+$ be contractions, with a rich, and let $c \in \tilde{C}$ be a contraction. There are positive contractions $p_1 \in \text{Her}(a)$ and $q_1 \in \text{Her}(y)$ such that p_1 is rich,*

$$\begin{aligned} \|\pi(a)\| > 1/2 &\implies \|\pi(p_1)\| = 1, \\ \|\pi(y)\| > 1/2 &\implies \|\pi(q_1)\| = 1 \end{aligned} \quad \text{for every representation } \pi, \quad (6.10)$$

and $\|e_{p_1} c e_{q_1}\|_{C^{**}} < 1$.

Proof. Choose ℓ from Lemma 6.2, set $k = \ell n^2$, and choose graph coefficients $\gamma_g, \sigma_g > 0$ with $\gamma_g^2 + k\sigma_g^2 < 1$. With these choices fixed, apply Proposition 3.1 to obtain its measurements X, B_i . Group the B_i into ℓ matrices $B_j = [B_{j,pq}] \in M_n(C)$, and choose the corresponding $S_{j,pq}$ by Proposition 6.3. The graph gives a rich positive contraction $p \in \text{Her}(a)$ with $\|\pi(p)\| = 1$ whenever $\|\pi(a)\| > 1/2$. If $d = y(1 - c^* p^2 c)y$ vanishes in a representation where $\|\pi(y)\| > s$, the graph would give all of (6.6), contrary to the choice of the entries. Thus

$$\pi(y(1 - c^* p^2 c)y) = 0 \implies \|\pi(y)\| \leq s = 1/4 \quad (6.11)$$

for every representation. Lemma 3.2 applies to this p , its norm signal, and the displayed implication, and gives the asserted p_1, q_1 . The one fixed n controlled both the number of entries and the algebra $M_n(C)$ in the quotient argument; only the finite dimensions N_r varied. $\square$

6.3 Local rich elements and their merger

The shrinking proposition will let us join rich elements once they are available. For $u \in C_+$, we first produce such an element inside $\text{Her}(u)$ at each irreducible representation that detects u . This local construction needs only the fixed-amplification hypothesis (F_n) ; nuclearity will enter when the elements are joined.

Lemma 6.5 (A rich element detected by one representation). *Let C be a C^* -algebra, and suppose that one integer $n \geq 1$ makes $h^{\oplus n}$ properly infinite for every $h \in C_+$. If $u \in C_+$ and an irreducible representation $\pi : C \rightarrow B(\mathcal{H})$ satisfies $\pi(u) \neq 0$, then there is a rich positive contraction $p \in \text{Her}(u)$ with $\|\pi(p)\| = 1$.*

Proof. We will find n orthogonal targets in $\text{Her}(u)$, each detected by π , and one positive element comparing into every target. Transporting n copies of its cutoff will allow (F_n) to produce richness inside $\text{Her}(u)$.

Put $\mathcal{H}_u = \overline{\pi(u)\mathcal{H}}$. This space is infinite dimensional. Otherwise $\pi(u)$ would have finite rank $r > 0$. Applying the stabilized representation to proper infiniteness of $u^{\oplus n}$ would approximate $\pi(u)^{\oplus 2n}$, of rank $2nr$, by positive operators of rank at most nr . On the $2nr$ -dimensional range of

the target, each such approximant has a unit vector in its kernel. Its distance from the target is therefore at least the least positive eigenvalue of $\pi(u)$, contradicting norm approximation.

The action of $D = \text{Her}(u)$ on $\mathcal{H}_u$ is irreducible. Indeed, $\pi(u)$ has no kernel on its support, so $\pi(u)\xi \neq 0$ for $0 \neq \xi \in \mathcal{H}_u$. Irreducibility of π gives $\pi(C)\pi(u)\xi = \mathcal{H}$. Applying $\pi(u)$ and taking closure yields

$$\overline{\pi(uCu)\xi} = \mathcal{H}_u.$$

Thus every nonzero vector is cyclic for the restricted hereditary algebra, which proves irreducibility.

Apply the selfadjoint form of Kadison transitivity [9, Theorem 1] to the irreducible unitization of this action. Choose $n + 1$ orthonormal vectors and prescribe $n + 1$ distinct real eigenvalues. A selfadjoint interpolant has the form $h_0 + \lambda 1$, with $h_0 \in D_{\text{sa}}$. The corresponding eigenvalues of $\pi(h_0)$ remain distinct, so at most one is zero. Choose n nonzero ones. Nonnegative continuous functions supported in pairwise disjoint neighborhoods of these values, avoiding zero and positive at the chosen values, give orthogonal elements

$$b_1, \dots, b_n \in D_+, \quad \pi(b_i) \neq 0.$$

The functions vanish at zero, so these elements belong to D also when D is nonunital.

There is an element $b' \in D_+$, still detected by π , with $b' \precsim b_i$ for every i . Start with $e' = b_1$. Given the current e' and the next b_i , choose a vector on which $\pi(e')$ is nonzero. Cyclicity for D and nonzero $\pi(b_i)$ give $h \in D$ such that $\pi(z) \neq 0$, where $z = b_i h e'$. Then

$$z^* z \in \text{Her}(e'), \quad z z^* \in \text{Her}(b_i), \quad z^* z \sim_{\text{Cu}} z z^*.$$

Replacing e' by $z^* z$ preserves nonzero image and every previous comparison, by transitivity. At the end we obtain b' and

$$(b')^{\oplus n} \precsim b_1 \oplus \dots \oplus b_n \sim_{\text{Cu}} \sum_{i=1}^n b_i \precsim u. \quad (6.12)$$

The middle equivalence uses orthogonality, and the last comparison uses $\sum_i b_i \in \text{Her}(u)$.

Choose $\tau > 0$ with $2\tau < \|\pi(b')\|$. Transport the n source copies in (6.12), after cutoff τ , into $\text{Her}(u)$. Let t be a ramp of b' , zero on $[0, \tau]$ and one on $[2\tau, \infty)$. It belongs to the common cutoff hereditary algebra and has represented norm one. The sum p of its n orthogonal transported copies is a positive contraction in $\text{Her}(u)$ with $\|\pi(p)\| = 1$. By (2.7), for every $\varepsilon \geq 0$,

$$(p - \varepsilon)_+ \sim_{\text{Cu}} ((t - \varepsilon)_+)^{\oplus n}.$$

The right side is properly infinite by the hypothesis applied to the scalar positive $(t - \varepsilon)_+$. Thus p is rich. $\square$

We now join two rich elements while retaining the union of their norm signals. At each step Proposition 6.4 is used with $c = 1$ in the unitization. Its support bound is one global norm below one for that step; the numerical bound may change at the next step.

Lemma 6.6 (Merging two rich elements). *Let C be nuclear and satisfy (F_n) for a fixed $n \geq 1$. For $u \in C_+$ and rich positive contractions $a, b \in \text{Her}(u)$, there is a rich positive contraction $c_0 \in \text{Her}(u)$ such that every representation π satisfies*

$$(\|\pi(a)\| > 1/2 \text{ or } \|\pi(b)\| > 1/2) \implies \|\pi(c_0)\| = 1. \quad (6.13)$$

Proof. All comparisons remain in $C \otimes \mathcal{K}$, and all uses of shrinking are in C . Since b is properly infinite, $b^{\oplus n} \precsim b$. Apply Lemma 2.2 with cutoff $1/8$ to obtain orthogonal copy maps

$$\theta_i(h) = w_i h w_i^* : \text{Her}((b - 1/8)_+) \longrightarrow \text{Her}(b) \subset \text{Her}(u) \quad (1 \leq i \leq n).$$

Let y_0 be a ramp of b , zero on $[0, 1/4]$ and one on $[1/2, 1]$, and set $a_0 = a$. For $1 \leq j \leq n$ we construct positive contractions a_j, y_j such that a_j is rich and belongs to $\text{Her}(u)$, y_j belongs to the common domain of the copy maps, and the following invariants hold:

$$\begin{aligned} a_j \theta_i(y_j) &= 0 \quad (1 \leq i \leq j), \\ \|\pi(a)\| > 1/2 &\implies \|\pi(a_j)\| = 1, \\ \|\pi(b)\| > 1/2 &\implies \|\pi(y_j)\| = 1 \quad \text{for every representation } \pi. \end{aligned} \tag{6.14}$$

Once the induction reaches $j = n$, we will set $c_0 = a_n + \sum_{i=1}^n \theta_i(y_n)$. The induction will make the summands orthogonal, and the final use of (F_n) will prove the sum of the n copies rich without requiring y_n itself to be rich.

Suppose a_{j-1}, y_{j-1} have been constructed. At $j = 1$, use the initial pair a_0, y_0 ; at later stages use the preceding invariants. Apply Proposition 6.4 to $a_{j-1}, \theta_j(y_{j-1})$, and $1 \in \tilde{C}$, and call its outputs p_j, q_j . Thus p_j is rich and belongs to $\text{Her}(a_{j-1})$, while

$$\kappa_j = \|e_{p_j} e_{q_j}\| < 1.$$

The hereditary-image equality (2.4) supplies a positive contraction $z_j \in \text{Her}(y_{j-1})$ such that

$$q_j = \theta_j(z_j). \tag{6.15}$$

This pullback is an algebra element. Its represented norms equal those of q_j by (2.5).

For $0 \leq t \leq 1$, put $f(t) = (2t - 1)_+$ and $g(t) = \min\{1, 2t\}$. We next move p_j away from a cutoff of q_j . The support bound ensures that multiplication by $1 - g(q_j)$ is bounded below on the support of p_j . The polar part of $(1 - g(q_j))p_j$ will therefore transport p_j , preserving its cutoff Cuntz classes and represented norms. Define

$$y_j = f(z_j), \quad H_j = 1 - g(q_j), \quad T_j = H_j p_j.$$

Here $T_j = p_j - g(q_j)p_j \in \text{Her}(u)$. In the universal representation, $g(q_j)$ is supported under e_{q_j} . For $\xi \in e_{p_j} \mathcal{H}$, this gives

$$\|H_j \xi\|^2 \geq \|(1 - e_{q_j}) H_j \xi\|^2 = \|(1 - e_{q_j}) \xi\|^2 \geq (1 - \kappa_j^2) \|\xi\|^2.$$

Since H_j is a contraction, we obtain the order bounds

$$(1 - \kappa_j^2) p_j^2 \leq T_j^* T_j \leq p_j^2. \tag{6.16}$$

Let $T_j = l_j |T_j|$ be its polar decomposition in the bidual. The order bounds give $\text{Her}(|T_j|) = \text{Her}(p_j)$ and $l_j^* l_j = e_{p_j}$. The polar membership argument in the proof of Lemma 2.2, applied now to T_j , gives

$$a_j = l_j p_j l_j^* \in \text{Her}(T_j T_j^*) \subset \text{Her}(u).$$

It is a positive contraction. The same argument makes each transported cutoff Cuntz equivalent to the corresponding cutoff of p_j , so a_j is rich. The normal extensions of the representations preserve $l_j^* l_j = e_{p_j}$; conjugating back by l_j^* shows that a_j and p_j have equal norm in every representation.

We check that this change preserves every earlier orthogonality relation. Since $gf = f$, functional calculus and (6.15) give

$$\theta_j(y_j)H_j = f(q_j)(1 - g(q_j)) = 0,$$

and hence $\theta_j(y_j)T_j = 0$. For $i < j$, the previous orthogonality and $p_j \in \text{Her}(a_{j-1})$ give $\theta_i(y_{j-1})p_j = 0$, while orthogonality of the copy ranges gives $\theta_i(y_{j-1})g(q_j) = 0$. Thus $\theta_i(y_{j-1})T_j = 0$. Each of these positive elements annihilates $\text{ran}(T_j)$, whose projection is $l_j l_j^*$, and so annihilates a_j . Because $y_j \in \text{Her}(y_{j-1})$, the same is true of $\theta_i(y_j)$ for $i < j$. This proves the orthogonality in (6.14).

The norm signals also persist simultaneously. Wherever $\|\pi(a)\| > 1/2$, the first shrinking step gives $\|\pi(p_1)\| = 1$, hence $\|\pi(a_1)\| = 1$; that signal is above $1/2$ and survives every subsequent step. Wherever $\|\pi(b)\| > 1/2$, the initial ramp has $\|\pi(y_0)\| = 1$. If $\|\pi(y_{j-1})\| = 1$, its copy under θ_j has norm one, shrinking gives $\|\pi(q_j)\| = 1$, and transport gives $\|\pi(z_j)\| = 1$. Since $f(1) = 1$, spectral mapping gives $\|\pi(y_j)\| = 1$.

Finally set

$$c_0 = a_n + \sum_{i=1}^n \theta_i(y_n).$$

All summands are orthogonal positive contractions in $\text{Her}(u)$. Their sum is a contraction, and in each representation its norm is the maximum of the summand norms. The preceding signals prove (6.13). For every $\tau \geq 0$, orthogonal functional calculus and (2.7) give

$$(c_0 - \tau)_+ \sim_{\text{Cu}} (a_n - \tau)_+ \oplus ((y_n - \tau)_+)^{\oplus n}.$$

The first summand is properly infinite because a_n is rich. The second is properly infinite by (F_n) , applied to the scalar positive $(y_n - \tau)_+$. Their block sum is properly infinite, so c_0 is rich. The construction includes zero inputs through zero copy maps and zero cutoffs. $\square$

6.4 Finite ideal assembly and absorption

Lemma 6.5 supplies a rich element wherever u is detected, and Lemma 6.6 joins any finite family while retaining all of its norm signals. These two facts put $(u - \varepsilon)_+$, for every $\varepsilon > 0$, in the ideal of a single rich element. Ideal comparison then supplies the doubled cutoff comparison needed for Theorem 6.1.

Proof of Theorem 6.1. Fix $u \in C_+$, and let $\mathcal{R}_u$ be the set of rich positive contractions in $\text{Her}(u)$. Let J be the closed ideal generated by all $(p - 1/2)_+$ with $p \in \mathcal{R}_u$. We claim that $u \in J$. Otherwise an irreducible representation of C/J would detect $u + J$. Its composition with the quotient map is an irreducible representation of C , so Lemma 6.5 gives $p \in \mathcal{R}_u$ with represented norm one. Then $(p - 1/2)_+$ has nonzero image, contrary to its membership in J .

The ideals generated by finite subfamilies of these cutoffs form an increasing directed family with dense union in J . For $\varepsilon > 0$, Lemma 2.3(i) therefore puts $(u - \varepsilon)_+$ in the ideal generated by $(p_i - 1/2)_+$ for finitely many $p_1, \dots, p_r \in \mathcal{R}_u$. Starting from the rich zero element, merge them successively by Lemma 6.6. The result is a rich $p \in \text{Her}(u)$ for which every representation satisfies

$$\|\pi(p_i)\| > 1/2 \text{ for some } i \implies \|\pi(p)\| = 1.$$

A norm-one signal survives the next merger because it is above $1/2$. The primitive ideal test in Lemma 2.3(ii) now gives $(p_i - 1/2)_+ \in \text{Id}(p)$ for every i . Consequently

$$(u - \varepsilon)_+ \in \text{Id}(p). \tag{6.17}$$

If the cutoff is zero, the zero element already suffices.

Since p is properly infinite, ideal comparison with this target in the stabilization, followed by hereditary comparison, yields

$$((u - \varepsilon)_+)^{\oplus 2} \precsim p \precsim u. \quad (6.18)$$

The target u is fixed as $\varepsilon \downarrow 0$. Explicitly, choose $\varepsilon_j \downarrow 0$ and, for each j , a comparison coefficient in (6.18) with error less than ε_j to the doubled cutoff. That cutoff differs from $u \oplus u$ by at most ε_j . The resulting coefficients therefore compare $u \oplus u$ to u in norm. Thus u is properly infinite. The fixed n was applied only to scalar positives and their scalar cutoffs throughout the proof. $\square$

Remark 6.7 (Extension of the tuple proof to exact algebras). The amplification-removal argument uses nuclearity only to obtain exactness of $E = \Psi(M_n(C))$ in Proposition 6.3. Quotients of exact C^* -algebras are exact, including in the nonseparable case [2, IV.3.4.20]. Finite matrix amplification also preserves exactness, since its tensor products and ideal kernels are the corresponding finite matrices over those for C . This deeper quotient-permanence theorem therefore makes the same avoidance, shrinking, merger, and ideal assembly valid when C is merely exact. It gives another proof of the amplification-removal conclusion of Theorem 1.2. The perturbation and localization proof below obtains that conclusion without invoking quotient permanence of exactness.

Proof of Corollary 1.3. Set $C = A$, the separable nuclear algebra in the corollary, with its stated fixed-amplification hypothesis. Theorem 6.1 gives (P), and Theorem 1.1 gives strong pure infiniteness. The general nonunital absorption implication is recorded in the paragraph following [13, Definition 3.1]. We spell out the central-sequence interface that includes the nonunital case.

Let C_ω be the norm ultrapower of C along a free ultrafilter on $\mathbb{N}$, with C embedded diagonally, and put

$$F(C) = \frac{C_\omega \cap C'}{\text{Ann}(C, C_\omega)}, \quad \text{Ann}(C, C_\omega) = \{x \in C_\omega : xC = Cx = 0\}.$$

Suppose first that $C \neq 0$. Separability makes C σ -unital, so $F(C)$ is unital by [10, Corollary 1.10(1)]. The central-sequence embedding statement immediately before [10, Proposition 4.4] gives a unital embedding $\mathcal{O}_\infty \hookrightarrow F(C)$: its ambient original algebra is the strongly purely infinite C , and the diagonal copy of C is the required separable nuclear subalgebra of C_ω . The algebra $\mathcal{O}_\infty$ is unital and separable, and Remark 4.6 supplies self-absorption and connectedness of its unitary group. The “in particular” clause of Proposition 4.4(4) therefore gives stable $\mathcal{O}_\infty$ -absorption. Since $U(\mathcal{O}_\infty) = U_0(\mathcal{O}_\infty)$, where U_0 is the identity component, the commutator subgroup of $U(\mathcal{O}_\infty)$ lies in $U_0(\mathcal{O}_\infty)$; part (5) therefore removes stabilization. Hence $C \cong C \otimes_{\min} \mathcal{O}_\infty$. The Proposition permits C to be nonunital; the unital embedding is into the annihilator quotient $F(C)$. If $C = 0$, the asserted isomorphism is immediate. $\square$

7 Uniform perturbation from exactness

We now begin the exact proof that (F_n) implies (P). Its first step is a perturbation estimate that applies simultaneously to every primitive quotient in which a fixed positive element has norm at least a prescribed positive threshold. The constant in that estimate comes from the following obstruction: there is one positive error threshold, independent of the unital exact algebra and the matrix size, below which a contraction cannot contain approximately reducing copies of every finite matrix contraction.

7.1 The universal contraction obstruction

Recall that a C^* -algebra B is *exact* if, for every closed two-sided ideal I of every C^* -algebra C , the canonical map

$$(B \otimes_{\min} C)/(B \otimes_{\min} I) \longrightarrow B \otimes_{\min} (C/I)$$

is an isomorphism. Here and below, homomorphisms of C^* -algebras are $*$ -homomorphisms. In the obstruction, the same finite-dimensional copy must approximately intertwine both a contraction and its adjoint.

Proposition 7.1 (Uniform gap). *There is a constant $\gamma > 0$ such that no unital exact C^* -algebra B contains a contraction v with the following property: for every integer $d \geq 1$ and every contraction $D \in M_d(\mathbb{C})$, there are a unital representation $\pi : B \rightarrow B(H_\pi)$ and an isometry $J : \mathbb{C}^d \rightarrow H_\pi$ satisfying*

$$\|\pi(v)J - JD\| < \gamma, \quad \|\pi(v)^*J - JD^*\| < \gamma. \quad (7.1)$$

The representation and the isometry are allowed to depend on d and D .

The proof has two parts. The universal unital algebra of a contraction fails exactness, as witnessed by a tensor quotient. If exact targets admitted the proposed reducing copies with arbitrarily small common error, their evaluations would give a lower bound on every matrix norm of each fixed polynomial space. Holding that space and one evaluation fixed while an ideal approximate unit varies would force the tensor quotient map to be injective. The two relations in (7.1) are what allow this argument to test words containing both the generator and its adjoint.

Let $\mathcal{U} = C^*(1, u)$ be the universal unital C^* -algebra generated by a contraction. Thus every contraction T on a Hilbert space determines a unital homomorphism $u \mapsto T$. One construction is to complete the unital $*$ -algebra of polynomials in u, u^* in the supremum seminorm over all such evaluations, after dividing out its null space. The seminorm of each polynomial is finite because a word in a contraction and its adjoint has norm at most one.

We shall use the following elementary quotient-norm formula. If D is unital and (e_λ) is a positive contractive approximate unit of an ideal $I \subset C$, then, for $z \in D \otimes_{\min} C$,

$$\|z + D \otimes_{\min} I\| = \lim_{\lambda} \|z(1 \otimes (1 - e_\lambda))\|. \quad (7.2)$$

When C is nonunital, the multiplication is performed in its unitization; its result still belongs to $D \otimes_{\min} C$. Indeed, $1 \otimes e_\lambda$ is an approximate unit for $D \otimes_{\min} I$. The product on the right represents the same quotient class as z , so its norm bounds the left side from below. For any $y \in D \otimes_{\min} I$, contractivity gives

$$\limsup_{\lambda} \|z(1 \otimes (1 - e_\lambda))\| \leq \|z - y\| + \limsup_{\lambda} \|y(1 \otimes (1 - e_\lambda))\| = \|z - y\|.$$

Taking the infimum over y proves the formula, including existence of the limit.

The formula also gives the form of subalgebra permanence that we need: exactness passes to unital C^* -subalgebras; see also [2, IV.3.4.3]. For a unital inclusion $D \subset B$, the spatial tensor norms on $D \otimes C$ and $D \otimes (C/I)$ are the norms inherited from the corresponding tensor products with B . Formula (7.2) therefore gives, for $z \in D \otimes_{\min} C$,

$$\|z + D \otimes_{\min} I\| = \|z + B \otimes_{\min} I\|.$$

If B is exact, this equals $\|(\text{id}_D \otimes q)z\|$, where $q : C \rightarrow C/I$ is the quotient map. This proves exactness of D .

Lemma 7.2. *The universal contraction algebra $\mathcal{U}$ is not exact.*

Proof. We first embed the full group algebra $Q = C^*(\mathbb{F}_2)$ into $\mathcal{U}$. The two free generators are sent to

$$\exp(2\pi i \operatorname{Re} u), \quad \exp(2\pi i \operatorname{Im} u).$$

To see that the resulting homomorphism is injective, take any pair of unitaries S, T on a Hilbert space. Bounded Borel functional calculus gives selfadjoint contractions H, K such that $S = \exp(\pi i H)$ and $T = \exp(\pi i K)$. The contraction $(H + iK)/2$ evaluates the two displayed unitaries to S, T . Thus every unital representation of Q is a restriction of an evaluation of $\mathcal{U}$, and the homomorphism is isometric. The logarithms need only belong to the bounded operators on the representation space. Courtney and Sherman give a related embedding with a different normalization in [6, Theorem 6.10].

Nonexactness of $C^*(\mathbb{F}_2)$ was established by Wassermann [23]. Here is an explicit proof for Q . We construct a representation of $Q \otimes_{\min} Q$ that annihilates $Q \otimes_{\min} I$, where I is the kernel of the regular quotient. Exactness would make it descend to $Q \otimes_{\min} (Q/I)$. A concrete tensor sum, however, has spatial norm in that algebra smaller than the norm the proposed descended representation would assign to it. This will exhibit the required failure of the tensor quotient map. Put $G = \mathbb{F}_2$, and write u_s for its canonical unitaries in Q . On $\ell^2(G)$ let

$$\lambda_s \delta_g = \delta_{sg}, \quad \rho_s \delta_g = \delta_{gs^{-1}}.$$

There is a state φ on $Q \otimes_{\min} Q$ with

$$\varphi(u_s \otimes u_t) = \begin{cases} 1, & s = t, \\ 0, & s \neq t. \end{cases} \quad (7.3)$$

For completeness, finite quotients supplying this state can be constructed directly. On the finite word ball of radius m in G , partially defined left translation by each of the two generators is an injection between subsets of equal size, hence extends to a permutation of the ball. Freeness gives a homomorphism q_m from G to the finite permutation group generated by these extensions. The inverse of each extension agrees with inverse left translation whenever both endpoints lie in the ball. If a reduced word has length at most m , applying its permutations to the identity follows its successive suffixes inside the ball and ends at that word. Consequently q_m distinguishes distinct elements of the word ball of radius m .

For the finite group $F_m = q_m(G)$, its commuting left and right regular representations give a representation of $C^*(F_m) \otimes_{\min} C^*(F_m)$: these algebras are finite direct sums of matrix algebras, for which the spatial and maximal tensor norms agree. Pulling its vector state at the identity back to $Q \otimes_{\min} Q$ gives the coefficient $\mathbf{1}_{q_m(s)=q_m(t)}$ on $u_s \otimes u_t$. These coefficients tend to those in (7.3). The group tensors span a dense $*$ -subalgebra, and all the states have norm one, so their limit on that span extends to a state on the spatial completion.

The GNS representation of φ is precisely

$$u_s \otimes u_t \longmapsto \lambda_s \rho_t \quad \text{on } \ell^2(G), \quad (7.4)$$

with cyclic vector δ_e . Indeed the cyclic vector associated with $u_s \otimes u_t$ corresponds to $\delta_{st^{-1}}$; the inner products agree because

$$s^{-1}a = t^{-1}b \iff st^{-1} = ab^{-1},$$

and left multiplication by $u_x \otimes u_y$ becomes $\lambda_x \rho_y$. These vectors span $\ell^2(G)$.

Let $I = \ker(Q \rightarrow C_r^*(G))$. The linear unitary $R\delta_g = \delta_{g^{-1}}$ satisfies $R\lambda_s R^* = \rho_s$. Thus the representation in (7.4) annihilates $Q \otimes_{\min} I$. Exactness of Q would make it descend to a representation of $Q \otimes_{\min} C_r^*(G)$, in which every $u_s \otimes \lambda_s$ fixes δ_e .

This contradicts a spatial norm bound. In a faithful unital representation $\pi : Q \rightarrow B(H)$, the unitary

$$V(\xi \otimes \delta_g) = \pi(u_g)\xi \otimes \delta_g$$

satisfies $V(1 \otimes \lambda_s)V^* = \pi(u_s) \otimes \lambda_s$. If a, b are the free generators, the norm of

$$u_a \otimes \lambda_a + u_{a^{-1}} \otimes \lambda_{a^{-1}} + u_b \otimes \lambda_b + u_{b^{-1}} \otimes \lambda_{b^{-1}}$$

in this faithful spatial representation is therefore the norm of the adjacency operator on the 4-regular tree. That norm is at most $2\sqrt{3} < 4$. To check the bound, orient each edge away from the identity, and for a finitely supported vector ξ bound the contribution of a parent-child edge by

$$2|\xi(p)\xi(c)| \leq 3^{-1/2}|\xi(p)|^2 + 3^{1/2}|\xi(c)|^2.$$

Every nonroot vertex has three children and one parent, and hence total coefficient $2\sqrt{3}$; the root has coefficient $4/\sqrt{3} \leq 2\sqrt{3}$. The quadratic-form bound extends by density and proves the asserted operator norm bound. In the purported descended representation the same sum has eigenvalue 4 on δ_e , which is impossible for a *-representation. Thus Q is not exact. Its unital inclusion in $\mathcal{U}$ and the inheritance proved above show that $\mathcal{U}$ is not exact either. $\square$

We have obtained the algebraic obstruction. To turn it into a fixed error bound, we next test polynomial norms on finite contractions and control those tests at every matrix amplification. The latter uniformity will allow one polynomial model to be held fixed while an ideal approximate unit varies.

Lemma 7.3 (Finite matrix tests). *For every integer $r \geq 1$ and every $Z \in M_r(\mathcal{U})$ whose entries are *-polynomials in u, u^* ,*

$$\|Z\|_{M_r(\mathcal{U})} = \sup_{d \geq 1, D \in M_d(\mathbb{C}), \|D\| \leq 1} \|Z(D, D^*)\|. \quad (7.5)$$

Proof. Contractivity of evaluations proves one inequality. Every unital representation of $\mathcal{U}$ is evaluation at the image of u , so these evaluations also compute the matrix norm on the left. For the other inequality, evaluate at a contraction $T \in B(H)$, and take a unit vector $\xi = (\xi_1, \dots, \xi_r) \in H^r$ arbitrarily close to attaining the norm of $Z(T, T^*)$. If the entries of Z have degree at most m , let

$$H_0 = \text{span}\{w(T, T^*)\xi_j : |w| \leq m, 1 \leq j \leq r\},$$

where the empty word is included. This is a finite-dimensional nonzero space. Set $D = P_{H_0}T|_{H_0}$, so that $D^* = P_{H_0}T^*|_{H_0}$. Applying a word successively from the right never changes the relevant vectors under compression, since all its suffix vectors lie in H_0 . Hence

$$w(D, D^*)\xi_j = w(T, T^*)\xi_j \quad (|w| \leq m),$$

and $Z(D, D^*)\xi = Z(T, T^*)\xi$. Taking the supremum over evaluations and over these vectors proves (7.5). The scalar norm-test formula is proved in [6, Proposition 2.2]; the finite-word compression used here is the argument in the proof of [6, Theorem 3.1]. Including all components of the test vector gives the matrix-valued assertion needed here. $\square$

Lemma 7.4 (A bound at every amplification). *Let $E \subset \mathcal{U}$ be a finite-dimensional linear space consisting of $*$ -polynomials in u, u^* . There is a finite constant $C_E \geq 0$ with the following property. Suppose v is a contraction in a unital C^* -algebra B , and fix one number $\theta > 0$. Assume that for every integer $d \geq 1$ and every contraction $D \in M_d(\mathbb{C})$, there are a unital representation $\pi : B \rightarrow B(H_\pi)$ and an isometry $J : \mathbb{C}^d \rightarrow H_\pi$ satisfying the two inequalities in (7.1) with θ in place of γ . For the homomorphism $\Phi : \mathcal{U} \rightarrow B$ with $\Phi(u) = v$, one then has*

$$\|\Phi^{(r)}(Z)\| \geq (1 - C_E \theta) \|Z\| \quad (r \geq 1, Z \in M_r(E)). \quad (7.6)$$

Proof. If $E = 0$, take $C_E = 0$. Otherwise choose a basis $P_1, \dots, P_k$ of E , with coordinate functionals $f_1, \dots, f_k$, and choose polynomial representatives $P_i = \sum_w a_{iw} w$. Define

$$C_E = \sum_{i=1}^k \|f_i\| \sum_w |a_{iw}| |w|. \quad (7.7)$$

Constants are allowed in the representatives; the empty word has length zero. For $Z = \sum_i B_i \otimes P_i \in M_r(E)$ and scalar unit vectors $\alpha, \beta \in \mathbb{C}^r$,

$$|\alpha^* B_i \beta| = |f_i(\alpha^* Z \beta)| \leq \|f_i\| \|Z\|.$$

It follows that $\|B_i\| \leq \|f_i\| \|Z\|$ for every matrix size r .

Given a finite contraction D , choose π, J as in the hypothesis. Both $\pi(v)$ and D are contractions, and telescoping the two intertwining relations gives

$$\|w(\pi(v), \pi(v)^*)J - Jw(D, D^*)\| \leq |w|\theta$$

for every word w , including words in both letters. More explicitly, the difference for a word $A_1 \cdots A_m$ is the sum of m terms, each consisting of contractions on either side of one difference $\pi(v)J - JD$ or $\pi(v)^*J - JD^*$. Put $J_r = 1_{\mathbb{C}^r} \otimes J$. The coefficient bound and (7.7) yield

$$\|\pi^{(r)}(\Phi^{(r)}(Z))J_r - J_r Z(D, D^*)\| \leq C_E \theta \|Z\|.$$

Since J_r is an isometry and $\pi^{(r)}$ is contractive,

$$\|Z(D, D^*)\| \leq \|\Phi^{(r)}(Z)\| + C_E \theta \|Z\|.$$

Taking the supremum supplied by Lemma 7.3 proves (7.6). In particular, C_E depends on the fixed polynomial space and its chosen representatives, but on none of r, d, D, B, v, π, J . $\square$

Proof of Proposition 7.1. If the Proposition were false, for every integer $j \geq 1$ there would be a unital exact algebra B_j and a contraction $v_j \in B_j$ satisfying (7.1) with $1/j$ in place of γ . Let $\Phi_j : \mathcal{U} \rightarrow B_j$ send u to v_j . Fix a finite-dimensional polynomial space E . Before choosing any matrix amplification, tensor factor, compression, or ideal, choose one integer $j = j(E) \geq \max\{1, 2C_E\}$. Lemma 7.4 then gives

$$\|Z\| \leq 2\|\Phi_j^{(r)}(Z)\| \quad (r \geq 1, Z \in M_r(E)). \quad (7.8)$$

This same j works after tensoring with an arbitrary C^* -algebra C : for every $z \in E \odot C$,

$$\|z\| \leq 2\|(\Phi_j \otimes \text{id}_C)z\|, \quad (7.9)$$

where $\odot$ denotes the algebraic tensor product and all norms are spatial. The assertion is immediate if $C = 0$. Otherwise, represent C faithfully on a nonzero Hilbert space K , and use faithful representations of $\mathcal{U}$ and B_j for their respective spatial tensor products. Compress the C representation to

a nonzero finite-dimensional subspace $K_0 \subset K$. If $z = \sum_i P_i \otimes c_i$, its compression has coefficients $P_{K_0} c_i|_{K_0} \in M_d(\mathbb{C})$ and, after interchanging the tensor factors, belongs to $M_d(E)$. Inequality (7.8) bounds its norm by twice the norm of the corresponding compression on the B_j side, which is at most $2\|(\Phi_j \otimes \text{id}_C)z\|$. The norms of these compressions have supremum $\|z\|$: as K_0 increases, the projections $1 \otimes P_{K_0}$ converge strongly to the identity, and their two-sided compressions converge strongly to z . This proves (7.9).

Now fix any ideal $I \subset C$, with quotient map $q : C \rightarrow C/I$ and a positive contractive approximate unit (e_λ) of I . For $z \in E \odot C$, every $z(1 \otimes (1 - e_\lambda))$ still belongs to $E \odot C$. Consequently (7.9) applies to the entire net with the single index $j(E)$. Formula (7.2) and exactness of B_j give

$$\begin{aligned} \|z + \mathcal{U} \otimes_{\min} I\| &= \lim_{\lambda} \|z(1 \otimes (1 - e_\lambda))\| \\ &\leq 2 \lim_{\lambda} \|(\Phi_j \otimes \text{id}_C)(z)(1 \otimes (1 - e_\lambda))\| \\ &= 2\|(\Phi_j \otimes \text{id}_{C/I})((\text{id}_{\mathcal{U}} \otimes q)z)\| \\ &\leq 2\|(\text{id}_{\mathcal{U}} \otimes q)z\|. \end{aligned}$$

Finite sums of polynomial tensors are dense in $\mathcal{U} \otimes_{\min} C$. Although $j(E)$ may vary with the polynomial space, the resulting constant 2 does not. The last inequality therefore extends by continuity to every $z \in \mathcal{U} \otimes_{\min} C$. It says that the canonical surjective homomorphism

$$\kappa : (\mathcal{U} \otimes_{\min} C)/(\mathcal{U} \otimes_{\min} I) \longrightarrow \mathcal{U} \otimes_{\min} (C/I)$$

has zero kernel: if $\kappa(z + \mathcal{U} \otimes_{\min} I) = 0$, the quotient class has norm zero. An injective $*$ -homomorphism of C^* -algebras is isometric, so the fixed factor 2 suffices to prove that κ is an isomorphism. Since C, I were arbitrary, $\mathcal{U}$ would be exact, contrary to Lemma 7.2. $\square$

Proposition 7.1 therefore supplies a single $\gamma > 0$ for all unital exact targets and all matrix sizes. The remaining part of this section converts that obstruction into a perturbation with a quotient norm lower bound.

7.2 Uniform perturbation in primitive quotients

We now turn the uniform gap into a perturbation that works in many primitive quotients at once. The fixed amplification condition (F_n) supplies finite reducing copies on a spectral cut; the gap then prevents one fixed element from nearly cancelling every such copy. We first record the two consequences of the comparison and transport results in Section 2 that this argument uses.

Lemma 7.5 (Fixed-amplification ideal comparison). *Let C be a C^* -algebra, let $h \in C_+$, and let $n \geq 1$ be an integer such that $h^{\oplus n}$ is properly infinite. For every $x \in \text{Id}(h)_+$ and every integer $k \geq 1$,*

$$x^{\oplus k} \precsim h^{\oplus n}. \quad (7.10)$$

In particular, if C satisfies (F_n) , this conclusion holds for every $h \in C_+$ with the same n .

Proof. In the common stabilization,

$$\text{Id}_{C \otimes \mathcal{K}}(h^{\oplus n}) = \text{Id}_C(h) \otimes \mathcal{K}.$$

Thus $x^{\oplus k}$ belongs to the ideal generated by $h^{\oplus n}$. The proper-target ideal comparison in Lemma 2.1 gives (7.10). This argument takes place in the stabilization and therefore covers every finite rectangular corner and nonunital C . It also covers $h = 0$, when $x = 0$. $\square$

Lemma 7.6 (Exact implementation of a spectral cut). *Let C be a C^* -algebra, let $m, s \geq 1$, and suppose that $x \in M_m(C)_+$ and $h \in M_s(C)_+$ satisfy $x \lesssim h$. If $f: [0, \|x\|] \rightarrow [0, \infty)$ is continuous and vanishes on a neighborhood of zero, then there is $R \in M_{s,m}(C)$ such that*

$$R^*R = f(x), \quad RR^* \in \text{Her}(h), \quad \|R\|^2 = \|f(x)\|. \quad (7.11)$$

Proof. If $f(x) = 0$, take $R = 0$. Otherwise, choose $\delta > 0$ so that f vanishes on $[0, \delta]$, and choose $0 < \varepsilon < \delta$. Then $f(x)^{1/2} \in \text{Her}((x - \varepsilon)_+)$. Apply Lemma 2.2 to $x \lesssim h$ at this cutoff, and let $w \in M_{s,m}(C^{**})$ be its rectangular partial isometry. Its initial projection is the support of $(x - \varepsilon)_+$. The membership and support conclusions of that lemma therefore give

$$R = wf(x)^{1/2} \in M_{s,m}(C), \quad R^*R = f(x), \quad RR^* = wf(x)w^* \in \text{Her}(h).$$

The norm identity follows from $\|R\|^2 = \|R^*R\|$. Only the product R , rather than w itself, is asserted to belong to the rectangular algebra. $\square$

For $P \in \text{Prim}(C)$, a subscript P denotes the image in C/P , also at matrix levels. An irreducible representation with kernel P faithfully represents this quotient and computes its norm. The primitive ideal test in Lemma 2.3 will let us recognize ideal membership from these quotient images.

The uniform gap will be applied to the unitization of a matrix algebra over C . We record the two permanence facts that make this a valid exact target, also when C has no unit.

Lemma 7.7 (The exact target for matrix models). *If A is an exact C^* -algebra and $n \geq 1$, then $\widehat{M_n(A)}$ is exact.*

Proof. Finite matrix exactness follows, for every C^* -algebra D_0 , from the identifications

$$M_n(A) \otimes_{\min} D_0 = M_n(A \otimes_{\min} D_0),$$

which preserve the ideal kernels in the definition of exactness. For the unitization step, let A_0 be a nonunital exact algebra, let $I \triangleleft D_0$ be a closed two-sided ideal, and write $q_0: D_0 \rightarrow D_0/I$. The scalar character $\chi: A_0 \rightarrow \mathbb{C}$ gives, for every $z \in \tilde{A}_0 \otimes_{\min} D_0$, the bounded decomposition

$$z = z_0 + 1 \otimes d, \quad d = (\chi \otimes \text{id})(z), \quad z_0 \in A_0 \otimes_{\min} D_0.$$

Indeed, both projections are bounded, and the decomposition holds first on algebraic tensors and then on their completion. If $(\text{id} \otimes q_0)(z) = 0$, applying $\chi \otimes \text{id}$ gives $d \in I$. The isometric inclusion

$$A_0 \otimes_{\min} (D_0/I) \subset \tilde{A}_0 \otimes_{\min} (D_0/I)$$

then shows $(\text{id} \otimes q_0)(z_0) = 0$. Exactness of A_0 gives $z_0 \in A_0 \otimes_{\min} I$, hence $z \in \tilde{A}_0 \otimes_{\min} I$. If $M_n(A)$ is already unital, this second step is unnecessary. $\square$

Fix, once and for all, a constant

$$0 < \sigma < \min\{1, \gamma/3\}, \quad (7.12)$$

where γ is supplied by Proposition 7.1. This σ is independent of the algebra and of the fixed amplification integer.

Proposition 7.8 (Uniform perturbation). *Let C be an exact C^* -algebra satisfying (F_n) for one fixed integer $n \geq 1$. Let $p, q \in C$ be contractions and let $h, k, c \in C_+$ satisfy*

$$p^*ph = h, \quad qq^*k = k, \quad c^{\oplus n} \lesssim h, \quad c^{\oplus n} \lesssim k. \quad (7.13)$$

For every $T \in C$ and every $\eta, \alpha > 0$, there is a contraction $y \in C$ such that

$$\|(T + \eta pyq)_P\| \geq \sigma \eta \quad \text{for every } P \in \text{Prim}(C) \text{ with } \|c_P\| \geq \alpha. \quad (7.14)$$

If h, k belong to a hereditary subalgebra $\mathcal{D} \subset C$, then y may be chosen in $\mathcal{D}$. If, in addition or independently, p, q are positive and $pq = 0$, then y may be chosen selfadjoint. Both conclusions may be imposed simultaneously. The constant σ is independent of all the data in the Proposition, including n and the norms of T, h, k, c .

Proof. Write $K = \{P \in \text{Prim}(C) : \|c_P\| \geq \alpha\}$. If K is empty, $y = 0$ satisfies all conclusions. Suppose henceforth that K is nonempty. The contradiction will have three levels of choices. First we fix comparison rows and a single contraction in $M_n(C)$. Each finite matrix contraction then determines an admissible perturbation. If it fails, a primitive quotient witnessing that failure supplies an approximate reducing copy for that fixed contraction. All these copies therefore come from one element of the same exact algebra $\widetilde{M_n(C)}$.

Fixed comparison rows. Choose a continuous function $e : [0, \infty) \rightarrow [0, 1]$ that vanishes near zero and is one on $[\alpha/2, \infty)$. Apply Lemma 7.6 to the last two comparisons in (7.13). This gives rows $R, S \in M_{1,n}(C)$ with

$$R^*R = S^*S = E = e(c)^{\oplus n}, \quad RR^* \in \text{Her}(h), \quad SS^* \in \text{Her}(k), \quad \|R\|, \|S\| \leq 1. \quad (7.15)$$

The relation $p^*ph = h$, together with its adjoint, shows that left and right multiplication by p^*p are the identity on $\text{Her}(h)$. Hence $(1 - p^*p)RR^*(1 - p^*p) = 0$, which implies $(1 - p^*p)R = 0$. The same argument for S gives

$$p^*pR = R, \quad qq^*S = S. \quad (7.16)$$

At this point fix the single element

$$V = \eta^{-1}R^*p^*Tq^*S \in M_n(C). \quad (7.17)$$

The rows R, S and the element V will not depend on any finite matrix model. Fix now also the global contraction

$$\widehat{V} = V(\max\{1, |V|\})^{-1} \in M_n(C).$$

The inverse is evaluated in the unitization, so its product with V remains in $M_n(C)$.

Reducing copies of arbitrary finite contractions. Let $d \geq 1$ and let $D \in M_d(\mathbb{C})$ be a contraction. Put $c_1 = (c - \alpha/2)_+$. Condition (F_n) and Lemma 7.5, applied to the scalar element c_1 , give

$$c_1^{\oplus d} \lesssim c_1^{\oplus n}.$$

Choose a continuous $\varphi : [0, \infty) \rightarrow [0, 1]$, vanishing near zero and equal to one on $[\alpha/4, \infty)$, and put $e_1(c) = \varphi(c_1)$. Lemma 7.6 yields $L \in M_{n,d}(C)$ such that

$$L^*L = e_1(c)^{\oplus d}, \quad LL^* \in \text{Her}(c_1^{\oplus n}), \quad \|L\| \leq 1. \quad (7.18)$$

Because $e(c)c_1 = c_1$, multiplication by E is the identity on $\text{Her}(c_1^{\oplus n})$. The range-square argument used for (7.16) gives $EL = L$. Define

$$w = LDL^* \in M_n(C), \quad \|w\| \leq 1, \quad EwE = w. \quad (7.19)$$

Scalar matrix multiplication here requires no unit in C .

For each $P \in K$, choose an irreducible representation $\pi_P: C \rightarrow B(H_P)$ with kernel P , and use its matrix amplifications. Identify C/P with the image of the induced faithful representation on H_P , and make the same identification at rectangular matrix levels. The spectral subspace $1_{(3\alpha/4, \infty)}(\pi_P(c))H_P$ is nonzero, since $\|c_P\| \geq \alpha$. Choose a unit vector ξ_P in that subspace, so that $\pi_P(e_1(c))\xi_P = \xi_P$. Define

$$J_P: \mathbb{C}^d \longrightarrow H_P^{\oplus n}, \quad J_P \lambda = L_P(\lambda \otimes \xi_P).$$

Equation (7.18) proves $J_P^* J_P = 1_d$. Both D and D^* preserve $\mathbb{C}^d \otimes \xi_P$, and the same equation gives

$$w_P J_P = J_P D, \quad w_P^* J_P = J_P D^*. \quad (7.20)$$

Thus one copy reduces both w_P and its adjoint. No eigenvector at the norm of c_P , and no continuous choice of the vectors ξ_P , is required.

Admissible tests and the two optional conclusions. For the chosen d, D , take

$$y_0 = -RwS^* \in C, \quad \|y_0\| \leq 1.$$

If $h, k \in \mathcal{D}$, then $RR^*, SS^* \in \mathcal{D}$. For a positive contractive approximate unit (a_λ) of $\mathcal{D}$, the identity

$$\|(1 - a_\lambda)R\|^2 = \|(1 - a_\lambda)RR^*(1 - a_\lambda)\| \longrightarrow 0$$

and its analogue for S give $a_\lambda R w S^* a_\lambda \rightarrow R w S^*$ in norm. Each approximant belongs to $\mathcal{DCD} \subset \mathcal{D}$, so $y_0 \in \mathcal{D}$. This uses the range squares of the rows and makes no assertion that their individual entries belong to $\mathcal{D}$.

If $p, q \geq 0$ and $pq = 0$, then $p^2 R = R$ implies $pR = R$: multiply $(1 - p^2)R = 0$ by $(1 + p)^{-1}$ in the unitization. Likewise $qS = S$. It follows that $pS = qR = 0$ and $S^*R = 0$. Consequently $y_0^2 = 0$ and $py_0^*q = 0$. The positive elements $y_0 y_0^*$ and $y_0^* y_0$ are orthogonal, whence

$$\|y_0 + y_0^*\|^2 = \|y_0 y_0^* + y_0^* y_0\| = \|y_0\|^2 \leq 1.$$

We may therefore use $y = y_0 + y_0^*$ when selfadjointness is requested, and $y = y_0$ otherwise. In both cases $pyq = py_0 q$; hereditary membership is preserved if it is also requested.

One global contraction and the uniform gap. Suppose that no admissible y satisfies (7.14). For each d, D , choose a witness $P \in K$ for the test just constructed. Then

$$\|(T - \eta p R w S^* q)_P\| < \sigma \eta.$$

Multiplying by the contractive outer factors $R^* p^*$ and $q^* S$, and using (7.16), gives

$$\|(V - E w E)_P\| = \|(V - w)_P\| < \sigma. \quad (7.21)$$

Although P may depend on d, D , the element V is the one fixed in (7.17).

Functional calculus commutes with quotient maps for the fixed global clipping $\widehat{V}$. Hence in each witness quotient

$$\|V_P - \widehat{V}_P\| = \|(|V_P| - 1)_+\| = (\|V_P\| - 1)_+ < \sigma. \quad (7.22)$$

The last strict inequality follows from (7.21) and $\|w_P\| \leq 1$. The clipping identity follows by squaring the absolute value of $V_P - V_P \max\{1, |V_P|\}^{-1}$ and applying scalar functional calculus. Combining (7.21), (7.22), and (7.20) gives

$$\|\widehat{V}_P J_P - J_P D\| < 2\sigma < \gamma, \quad \|\widehat{V}_P^* J_P - J_P D^*\| < 2\sigma < \gamma. \quad (7.23)$$

By Lemma 7.7, $B = \widetilde{M_n(C)}$ is a unital exact algebra. The amplified representations $\pi_P^{(n)}$ extend unitally to B . Although the witnessing P varies with d, D , the contraction $\widehat{V}$ and the exact algebra B remain fixed. Thus $\widehat{V} \in B$ has, for every d, D , a unital representation and an isometry satisfying both inequalities in (7.23). This contradicts Proposition 7.1 and proves the Proposition with both optional conclusions and the same σ . $\square$

8 Exact localization and removal of amplification

We now use Proposition 7.8 to prove the fixed-amplification part of Theorem 1.2. Throughout this section, let C be an exact C^* -algebra satisfying (F_n) for one fixed integer $n \geq 1$, and retain the constant $0 < \sigma < 1$ from (7.12). For $P \in \text{Prim}(C)$, write $a_P = a + P$, with its norm computed in any irreducible representation having kernel P . A hereditary subalgebra $\mathcal{D} \subset C$ is nonzero at P when $\mathcal{D} \not\subset P$.

Our immediate objective is to construct a square-zero element in a hereditary subalgebra that is nonzero on a prescribed quasi-compact subset of $\text{Prim}(C)$. Its initial and range squares are orthogonal and Cuntz equivalent. Repeating the construction on one fixed norm slice will remove the amplification integer n and give (P). Theorem 1.1 will then supply the strong pure infiniteness conclusion. All finite covers in the construction concern the primitive ideal space; the partition of unity used below is on a real interval.

8.1 Local elements and finite covers

The next lemma uses only (F_n) : finite-dimensional hereditary images are excluded, and irreducibility then supplies common positive elements locally. Exactness enters later through Proposition 7.8, when these local choices must work simultaneously on a whole quasi-compact set.

Lemma 8.1 (Local common elements). *Let C be an exact C^* -algebra satisfying (F_n) for one fixed integer $n \geq 1$, and let $P \in \text{Prim}(C)$.*

- (i) *If $h \in C_+$ and $h_P \neq 0$, then for every integer $m \geq 1$ there are m pairwise orthogonal elements of $\text{Her}(h)_+$, each nonzero at P .*
- (ii) *If $h, k \in C_+$ are nonzero at P , there is $c \in C_+$ such that $c_P \neq 0$ and*

$$c^{\oplus n} \precsim h, \quad c^{\oplus n} \precsim k.$$

If h, k belong to a hereditary subalgebra $\mathcal{D}$, c can also be chosen in $\mathcal{D}$.

Proof. Fix an irreducible representation $\pi : C \rightarrow B(H_\pi)$ with kernel P . First, no nonzero $\pi(a)$, $a \in C_+$, has finite rank. Indeed, if its rank were $d \in \mathbb{N}$, applying the amplified representation to $a^{\oplus 2n} \precsim a^{\oplus n}$ would express an operator of rank $2nd$ as a norm limit of operators of rank at most nd . This is impossible, since the set of operators of rank at most a fixed finite integer is norm closed.

Put $H_h = \overline{\pi(h)H_\pi}$. We claim that the restriction of $\pi(\text{Her}(h))$ to H_h is strongly dense in $B(H_h)$. Irreducibility and the bicommutant theorem [2, I.9.1.1] give strong density of $\pi(C)$ in $B(H_\pi)$. Consequently the strong closure of $\pi(\text{Her}(h))$ contains $\pi(h)B(H_\pi)\pi(h)$. For $t > 0$, let $E_t = 1_{[t, \infty)}(\pi(h))$. Every operator $E_t T E_t$, with $T \in B(H_h)$ extended by zero on $H_h^\perp$, has the form $\pi(h)S\pi(h)$: the inverse of $\pi(h)$ on $E_t H_\pi$ is bounded. As $t \downarrow 0$, $E_t T E_t \rightarrow T$ strongly on H_h . This proves the claim.

The space H_h is infinite dimensional by the finite-rank obstruction. Choose orthonormal vectors $\xi_1, \dots, \xi_m$ in H_h and distinct numbers $\lambda_1, \dots, \lambda_m \in (0, 1)$. Let $T \in B(H_h)$ be the positive contraction

with $T\xi_j = \lambda_j\xi_j$ and zero on their orthogonal complement. Choose nonnegative continuous functions f_j on $\mathbb{R}$, with disjoint compact supports in $(0, 1)$ and $f_j(\lambda_j) = 1$. Kaplansky density [2, I.9.1.3] supplies a bounded net of selfadjoint elements b_λ of the represented hereditary algebra converging strongly to T . Continuous functional calculus on bounded selfadjoint sets is strongly continuous; this follows by uniform polynomial approximation on a common bounded interval. Hence $f_j(b_\lambda)\xi_j \rightarrow \xi_j$. One b_λ makes all these m vectors nonzero. Lift it to a selfadjoint $b \in \text{Her}(h)$. The elements $f_j(b)$ lie in $\text{Her}(h)$, are pairwise orthogonal, and are nonzero at P . The functions vanish at zero, so this argument also applies to a nonunital hereditary algebra. This proves (i).

For two positives u, v nonzero at P , strong density of $\pi(C)$ gives $x \in C$ with $\pi(uxv) \neq 0$. For example, choose a vector on which $\pi(v)$ is nonzero and a rank-one operator carrying its image to a vector not annihilated by $\pi(u)$; approximate that operator strongly on the image under $\pi(v)$. With $z = uxv$, the positive $d = zz^*$ satisfies

$$d_P \neq 0, \quad d \in \text{Her}(u), \quad d \sim_{\text{Cu}} z^*z \in \text{Her}(v).$$

Here and below $\sim_{\text{Cu}}$ denotes Cuntz equivalence, and the comparison rules are those of Lemma 2.1. For a finite list of positives nonzero at P , repeating this operation produces one positive, nonzero at P , subequivalent to every element of the list. At each step the new element lies in the hereditary algebra of the preceding one, so all previous comparisons and any prescribed hereditary membership persist.

Apply (i) with $m = n$ separately to h and k , obtaining orthogonal lists $h_1, \dots, h_n$ and $k_1, \dots, k_n$. Take a common element c for the resulting list of $2n$ positives. Then

$$c^{\oplus n} \lesssim \bigoplus_{j=1}^n h_j \sim_{\text{Cu}} \sum_{j=1}^n h_j \lesssim h,$$

and the same argument gives $c^{\oplus n} \lesssim k$. The asserted choice in $\mathcal{D}$ follows from the hereditary membership just observed. $\square$

We record the norm-attainment fact that ensures a strict inequality in the finite-cover argument.

Lemma 8.2 (Pure-state norm attainment). *If B is a C^* -algebra and $0 \neq b \in B_+$, there is an irreducible representation π of B with $\|\pi(b)\| = \|b\|$.*

Proof. In the state space of the unitization $\tilde{B}$, the set

$$F_b = \{\varphi : \varphi(b) = \|b\|\}$$

is nonempty: evaluation at the largest spectral value of b on $C^*(1, b)$ extends to a state of $\tilde{B}$ by the positive Hahn–Banach extension theorem [2, II.6.3.1]. It is a weak-star compact face, since every state has value at most $\|b\|$ on b . By the Krein–Milman theorem it has an extreme point, which is therefore a pure state of $\tilde{B}$; see [2, II.6.3.2–3]. Let $\tilde{\pi}$ be its irreducible GNS representation [2, II.6.4.8]. It does not annihilate B , because its value on b is positive. The closure of $\tilde{\pi}(B)H_{\tilde{\pi}}$ is a nonzero reducing subspace for $\tilde{\pi}(\tilde{B})$, hence the whole space. If an operator commutes with $\tilde{\pi}(B)$, then its commutator with $\tilde{\pi}(x)$ annihilates $\tilde{\pi}(B)H_{\tilde{\pi}}$ for every $x \in \tilde{B}$, since $xB \subset B$. It therefore commutes with $\tilde{\pi}(\tilde{B})$. Thus the restriction $\pi = \tilde{\pi}|_B$ is nondegenerate and irreducible. The vector state gives $\|\pi(b)\| \geq \|b\|$, and contractivity gives equality. $\square$

Lemma 8.3 (Primitive norm slices). *Let C be a C^* -algebra. For $c \in C_+$ and $\alpha > 0$, the set*

$$\Omega(c, \alpha) = \{P \in \text{Prim}(C) : \|c_P\| > \alpha\}$$

is open, and

$$K(c, \alpha) = \{P \in \text{Prim}(C) : \|c_P\| \geq \alpha\}$$

is quasi-compact. No Hausdorff hypothesis is required.

Proof. The first set is $\{P : (c - \alpha)_+ \notin P\}$, which is open in the hull-kernel topology. For the second assertion, extend any relative open cover to ambient open sets. Write those sets as

$$\mathcal{O}_\lambda = \{P : J_\lambda \not\subset P\}, \quad I = \overline{\sum_\lambda J_\lambda},$$

where each J_λ is a closed ideal of C . If $\|c + I\| \geq \alpha$, Lemma 8.2 applied in C/I gives an irreducible representation attaining that norm. Its pullback has a primitive kernel containing I , belongs to $K(c, \alpha)$, and lies outside every $\mathcal{O}_\lambda$. This contradicts the covering property. Thus $\|c + I\| < \alpha$; if $C/I = 0$, the same inequality is immediate. Choose $d \in I$ with $\|c - d\| < \alpha$, and then approximate d by an element $d_F \in \sum_{\lambda \in F} J_\lambda$ for a finite index set F , closely enough that $\|c - d_F\| < \alpha$ still holds. Any primitive ideal containing every J_λ , $\lambda \in F$, has $\|c_P\| < \alpha$. Hence the finitely many $\mathcal{O}_\lambda$, $\lambda \in F$, cover $K(c, \alpha)$. The proof uses quasi-compactness only and does not assert that $K(c, \alpha)$ is closed. $\square$

Lemma 8.4 (Finite coupling). *Let C be an exact C^* -algebra satisfying (F_n) for one fixed integer $n \geq 1$. Let $K_0 \subset \text{Prim}(C)$ be quasi-compact and let $\mathcal{D} \subset C$ be hereditary. Suppose that at each $P \in K_0$ one can choose positive contractions $p, q \in \mathcal{D}$ and $h, k \in \mathcal{D}_+$ satisfying*

$$pq = 0, \quad ph = h, \quad qk = k, \quad h_P \neq 0, \quad k_P \neq 0.$$

There are finitely many such choices (p_i, q_i, h_i, k_i) and one selfadjoint $U \in \mathcal{D}$ such that, at every $P \in K_0$,

$$(p_i U q_i)_P \neq 0$$

for at least one selected index i . Any support conditions in the original choices are retained in the selected ones.

Proof. If K_0 is empty, choose $U = 0$ and the empty family. Otherwise, Lemma 8.1 associates to every original choice a positive c nonzero at its specified point, with $c^{\oplus n} \precsim h, k$. Choose $0 < \alpha < \|c_P\|$. The resulting open sets $\Omega(c, \alpha)$ cover K_0 , so choose a finite subcover $\Omega_i = \Omega(c_i, \alpha_i)$, $1 \leq i \leq N$.

Set $\theta = \sigma/4$, $\eta_i = \theta^{i-1}$, and $U_0 = 0$. At stage i , apply Proposition 7.8 with $T = p_i U_{i-1} q_i$, scale η_i , and the selected data. The identities $p_i h_i = h_i$, $q_i k_i = k_i$ imply $p_i^2 h_i = h_i$, $q_i^2 k_i = k_i$. The simultaneous hereditary and selfadjoint options give a selfadjoint contraction $y_i \in \mathcal{D}$ such that $U_i = U_{i-1} + \eta_i y_i$ satisfies

$$\|(p_i U_i q_i)_P\| \geq \sigma \eta_i \quad (P \in K(c_i, \alpha_i)).$$

Let $U = U_N$. Later perturbations change the i th corner by at most

$$\sum_{j>i} \eta_j \leq \eta_i \frac{\theta}{1-\theta} = \eta_i \frac{\sigma}{4-\sigma} < \frac{\sigma \eta_i}{2}.$$

Consequently

$$\|(p_i U q_i)_P\| > \frac{\sigma \eta_i}{2} \quad (P \in K(c_i, \alpha_i)). \quad (8.1)$$

In particular each selected corner is nonzero throughout its entire open set Ω_i , proving the assertion. $\square$

When applying Lemma 8.4, it suffices to produce two orthogonal positives u, v nonzero at the specified point. Indeed, choose $0 < \beta < \|u_P\|$ and a continuous $f : [0, \infty) \rightarrow [0, 1]$ that is zero near zero and one on $[\beta, \infty)$. Then $p = f(u)$ and $h = (u - \beta)_+$ satisfy $ph = h$ and $h_P \neq 0$. The analogous construction from v gives q, k . All four elements retain the hereditary and spectral support restrictions of u, v .

8.2 A square-zero element on a quasi-compact set

Proposition 8.5. *Let C be an exact C^* -algebra satisfying (F_n) for one fixed integer $n \geq 1$. Let $K_0 \subset \text{Prim}(C)$ be quasi-compact, and let $\mathcal{D} \subset C$ be a hereditary subalgebra nonzero at every point of K_0 . There exists $x \in \mathcal{D}$ such that*

$$x^2 = 0, \quad x_P \neq 0 \quad (P \in K_0).$$

Proof. If $K_0 = \emptyset$, take $x = 0$. Assume henceforth that K_0 is nonempty. We first construct a selfadjoint element that is nonscalar on every relevant hereditary support space. Coupling its separated spectral windows and deleting the nearby spectral blocks will produce a selfadjoint element whose positive and negative parts are both nonzero at every point of K_0 . A final coupling of those two orthogonal parts gives the required square-zero element.

First coupling: a nonscalar selfadjoint element. At each $P \in K_0$, choose $h \in \mathcal{D}_+$ with $h_P \neq 0$. Lemma 8.1(i), with $m = 2$, provides two orthogonal positives in $\text{Her}(h) \subset \mathcal{D}$ nonzero at P . Apply the preceding cutoff construction to these two positives and use the resulting four inputs in Lemma 8.4 to obtain $U = U^* \in \mathcal{D}$. For an irreducible representation π with kernel P , put $H_{\mathcal{D}, \pi} = \pi(\mathcal{D})H_\pi$. This space reduces $\pi(\mathcal{D})$, which acts as zero on its orthogonal complement. At P one selected pUq is nonzero. Since $p, q \in \mathcal{D}$ and $pq = 0$, this implies

$$\pi(U)|_{H_{\mathcal{D}, \pi}} \text{ is not a scalar operator} \quad (P \in K_0). \quad (8.2)$$

In particular, nonscalarity is obtained on the support space of $\mathcal{D}$, without using its zero action on the complement.

Second coupling: separated spectral windows. At a specified $P \in K_0$, the selfadjoint operator in (8.2) has two distinct spectral values. Choose disjoint closed intervals $I, J \subset \mathbb{R}$ at positive distance, and continuous nonnegative functions F, G supported in I, J , respectively, such that both $F(\pi(U))|_{H_{\mathcal{D}, \pi}}$ and $G(\pi(U))|_{H_{\mathcal{D}, \pi}}$ are nonzero. The functions may be chosen on all of $\mathbb{R}$ with compact support. If a window contains zero, its function need not vanish at zero. We evaluate scalar functions in $\mathcal{D} + \mathbb{C}1$ inside a fixed unitization $\tilde{C}$ of C , and take spectral projections in $\tilde{C}^{**}$. Let (d_λ) be a positive contractive approximate identity of $\mathcal{D}$. The positive elements

$$F(U)d_\lambda F(U), \quad G(U)d_\lambda G(U)$$

belong to $\mathcal{D}$. Their images on $H_{\mathcal{D}, \pi}$ converge strongly to $F(\pi(U))^2$ and $G(\pi(U))^2$, respectively, so a choice of middle elements makes both images nonzero. These positives are supported, globally in the bidual, on $1_I(U)$ and $1_J(U)$.

The same cutoff construction gives the corresponding p, q, h, k , retaining these spectral supports. Apply Lemma 8.4 again to obtain $V = V^* \in \mathcal{D}$ and finitely many selected pairs (p_i, q_i) , supported in windows (I_i, J_i) , such that

$$p_i = 1_{I_i}(U)p_i 1_{I_i}(U), \quad q_i = 1_{J_i}(U)q_i 1_{J_i}(U), \quad (8.3)$$

for every $P \in K_0$ there is i with $(p_i V q_i)_P \neq 0$.

Put $\rho = \min_i \text{dist}(I_i, J_i) > 0$. The positive elements p_i, q_i need not commute with U ; only the displayed support identities will be used.

Spectral deletion produces both signs. We retain the corners in (8.3) while removing all sufficiently narrow diagonal spectral blocks. A positive operator with all those diagonal blocks zero must vanish; the same holds for a negative operator. Since the retained corners prevent vanishing, this will force both signs in every quotient over K_0 .

Fix a compact interval $[-M, M]$ containing the spectrum of U in the unitization. Take a finite continuous nonnegative partition of unity $f_1, \dots, f_s$ on this interval, whose supports S_j have diameter less than $\rho/8$. Define

$$H = \sum_{\text{dist}(S_j, S_k) > \rho/2} f_j(U) V f_k(U). \quad (8.4)$$

The summation uses ordered pairs, and its condition is symmetric in j, k , so $H = H^*$. Although a factor $f_j(U)$ may have a scalar part, it lies in $\mathcal{D} + \mathbb{C}1$; the middle factor $V \in \mathcal{D}$ implies $H \in \mathcal{D}$.

For each selected pair (p_i, q_i) we have

$$p_i H q_i = p_i V q_i. \quad (8.5)$$

Indeed, expand $p_i V q_i$ using $\sum_j f_j(U) = 1$ on both sides. For a summand $p_i f_j(U) V f_k(U) q_i$ to be nonzero, the supports S_j, S_k must meet I_i, J_i , respectively, by (8.3). Their distances then satisfy

$$\text{dist}(S_j, S_k) \geq \text{dist}(I_i, J_i) - \text{diam}(S_j) - \text{diam}(S_k) > \frac{3\rho}{4} > \frac{\rho}{2}.$$

Every potentially nonzero summand is therefore retained in (8.4). This argument requires no commutation of p_i, q_i with U . Equations (8.3) and (8.5) show that $H_P \neq 0$ for every $P \in K_0$.

Let B be any Borel interval of length less than $\rho/4$. The two supports in a retained pair cannot both meet B , since their distance exceeds $\rho/2$. Hence

$$1_B(U) H 1_B(U) = 0.$$

Partition $[-M, M]$ into finitely many disjoint Borel intervals B_j of length less than $\rho/4$, assigning endpoints so that their spectral projections sum to the identity. In any unital extension of a representation, the resulting projections E_j thus satisfy $E_j \pi(H) E_j = 0$ and $\sum_j E_j = 1$. If $\pi(H) \geq 0$, then

$$\|\pi(H)^{1/2} E_j \xi\|^2 = \langle E_j \pi(H) E_j \xi, \xi \rangle = 0$$

for every j, ξ , whence $\pi(H) = 0$. Applying the same argument to $-\pi(H)$ excludes a nonzero negative semidefinite image. The spectral projection calculations can equivalently be made in the bidual and passed through the normal extensions of the representations. We conclude that

$$(H_+)_P \neq 0, \quad (H_-)_P \neq 0 \quad (P \in K_0). \quad (8.6)$$

Both positive parts belong to $\mathcal{D}$ and are orthogonal.

Third coupling: the square-zero product. At every $P \in K_0$, apply the same cutoff construction to H_+, H_- to obtain $p, h \in \text{Her}(H_+)$ and $q, k \in \text{Her}(H_-)$ for Lemma 8.4. The contractions p, q are orthogonal. The resulting $U' = U'^* \in \mathcal{D}$ has a selected nonzero corner at every point of K_0 . Put

$$x = H_+ U' H_- \in \mathcal{D}.$$

If $x_P = 0$, then $(pU'q)_P = 0$ for every $p \in \text{Her}(H_+)$ and $q \in \text{Her}(H_-)$: approximate them in norm by elements H_+aH_+ and H_-bH_- , and use

$$(H_+aH_+)U'(H_-bH_-) = H_+a x b H_-.$$

This contradicts the selected nonzero corner at P . Thus $x_P \neq 0$ throughout K_0 , while $x^2 = H_+U'(H_-H_+)U'H_- = 0$. All constructed elements lie in $\mathcal{D}$; spectral projections have only been used to verify their supports. $\square$

8.3 Removing the amplification

We have constructed a square-zero element on every prescribed quasi-compact set where a hereditary subalgebra is nonzero. We now apply this result repeatedly on the same primitive norm slice. The resulting number of orthogonal equivalent copies can be made at least the single amplification integer in (F_n) .

Proof of Theorem 1.2. Fix $a \in C_+$ and $r > 0$. The set $K_0 = K(a, r)$ is quasi-compact by Lemma 8.3. If it is empty, every primitive image of $(a - r)_+$ is zero, so $(a - r)_+ = 0$, and $((a - r)_+)^{\oplus 2} \lesssim a$ is immediate. Suppose that K_0 is nonempty. Proposition 8.5 applied to $\text{Her}(a)$ gives $x \in \text{Her}(a)$ with $x^2 = 0$ and $x_P \neq 0$ throughout K_0 . Set $z_1 = x^*x$. Its Cuntz-equivalent positive xx^* is orthogonal to it, since $(x^*x)(xx^*) = x^*x^2x^* = 0$. Therefore

$$z_1^{\oplus 2} \sim_{\text{Cu}} z_1 \oplus xx^* \sim_{\text{Cu}} z_1 + xx^* \lesssim a, \quad (z_1)_P \neq 0 \quad (P \in K_0).$$

Repeat the construction inside $\text{Her}(z_1)$, and then inside successive hereditary subalgebras, always over the same K_0 . It yields $z_j \in C_+$ nonzero throughout K_0 and $z_{j+1}^{\oplus 2} \lesssim z_j$. Finite direct sums and transitivity give

$$z_j^{\oplus 2^j} \lesssim a.$$

Choose a finite $j \geq 1$ with $2^j \geq n$. Dropping extra summands gives $z_j^{\oplus n} \lesssim a$. If $(z_j)_P = 0$, then $P \notin K_0$, so $\|a_P\| < r$ and $((a - r)_+)_P = 0$. The primitive ideal test in Lemma 2.3 therefore gives $(a - r)_+ \in \text{Id}(z_j)$. Condition (F_n) makes $z_j^{\oplus n}$ properly infinite because z_j is a scalar positive element. Lemma 7.5, used with two copies on the left, then implies

$$((a - r)_+)^{\oplus 2} \lesssim z_j^{\oplus n} \lesssim a.$$

Only scalar positives $z_j \in C$ have been tested against (F_n) . Finally, $\|a - (a - r)_+\| \leq r$. As $r \downarrow 0$, norm-limit closure of comparison with the fixed target a gives

$$a \oplus a \lesssim a \quad (a \in C_+). \quad (8.7)$$

This includes $a = 0$; when $n = 1$, it also follows directly from the hypothesis. Therefore C satisfies (P). Theorem 1.1 now gives strong pure infiniteness, completing the proof. $\square$

The removal of amplification also gives the comparison form needed for the direct quantitative argument in Section 9. The following conclusion concerns the entire closed ideal, rather than only one cutoff or primitive norm slice.

Corollary 8.6 (Whole-ideal comparison). *Let C be a C^* -algebra satisfying (P). For every $h \in C_+$, $c \in \text{Id}(h)_+$, and integer $k \geq 1$,*

$$c^{\oplus k} \lesssim h.$$

If C is exact, Proposition 7.8 may be applied with multiplicity $n = 1$, with the same constant σ and all its stated options.

Proof. Property (P) makes h properly infinite, so Lemma 7.5 applies with multiplicity one. If $h = 0$, the ideal is zero and the assertion is immediate. For exact C , property (P) is precisely the amplification hypothesis of Proposition 7.8 with $n = 1$ for every scalar positive element. Its constant was fixed independently of n , so neither the constant nor the hereditary and selfadjoint options change. $\square$

9 Direct quantitative separation under exactness

We give a direct proof of strong pure infiniteness from the exact perturbation principle. Throughout this section, C is exact and satisfies (P). This is (F_n) with $n = 1$, so Proposition 7.8 is available with that multiplicity and its fixed constant $0 < \sigma < 1$. Corollary 8.6 applies to the whole ideal $\text{Id}(h)$ generated by each positive h . For an exact algebra satisfying (F_n) , the scalar conclusion (8.7) proved in Section 8 supplies (P) before that section invokes the general theorem.

The quantitative feature of this proof is a fixed lower norm bound in primitive quotients, retained while a mixed product tends to zero. We first construct positive elements in two prescribed hereditary subalgebras with this property. Spectral clipping then supplies bounded comparison coefficients, so the small product survives Cuntz implementation. The resulting estimates verify the strong-infiniteness inequalities directly. Thus this proof uses the comparison tools and the exact perturbation and localization results in Sections 2, 7, and 8.

Lemma 9.1 (Separation with a fixed quotient norm bound). *Let C be an exact C^* -algebra satisfying (P), and let $0 < \sigma < 1$ be the constant in Proposition 7.8 with $n = 1$. Let $a, b \in C_+$ be contractions, and let $0 < r < 1$. Set*

$$m = \frac{r\sigma^2}{1024}.$$

For every $0 < \ell < 1/10$ there are positives $Q_a \in \text{Her}(a)$ and $Q_b \in \text{Her}(b)$ such that

$$\begin{aligned} \|(Q_a)_P\| &\geq m & \text{if } \|a_P\| &\geq r, \\ \|(Q_b)_P\| &\geq m & \text{if } \|b_P\| &\geq r, \end{aligned} \tag{9.1}$$

$$\|Q_a Q_b\| \leq 20\sqrt{\ell}. \tag{9.2}$$

Here and below P ranges over $\text{Prim}(C)$.

Proof. Write the polar decomposition in C^{**} as

$$ab = vu, \quad u = |ab|, \quad u_* = |(ab)^*|.$$

All three positive elements a, b, u are contractions. Choose continuous functions on $[0, 1]$ to separate the small singular values of ab from a spectral corner on which an inverse square root is available. The small values will give small cross products; on the remaining corner we will preserve exact cancellation while perturbing the two sides separately. Use the following functions:

- $0 \leq L \leq 1$, $L = 1$ on $[0, 5\ell]$, and $L = 0$ on $[6\ell, 1]$; put $N = 1 - L$;
- $0 \leq g \leq 1$, $g = 0$ on $[0, \ell]$, and $g = 1$ on $[2\ell, 1]$;
- $0 \leq R \leq 1$, $R = 0$ on $[0, \ell/2]$, and $R(s) = \sqrt{\ell/s}$ for $s \geq \ell$.

Linear interpolation on the unspecified intervals gives such functions. A function letter without an argument denotes evaluation at u . L selects the small singular values, g confines the perturbations

away from zero, and R normalizes u on that region. Functional calculus and the polar decomposition give

$$L(u_*)v = vL(u), \quad N(u_*) = vNv^*, \quad gN = N, \quad (RuR - \ell)g = 0. \quad (9.3)$$

The second identity uses $N(0) = 0$; the first remains valid with $L(0) = 1$. Moreover

$$vR = ab \frac{R(u)}{u} \in C, \quad vN = ab \frac{N(u)}{u} \in C,$$

where both quotient functions are extended continuously by zero at $s = 0$, since R and N vanish near zero. We use the unitization for scalar terms throughout this proof.

The four columns and their products. We first specify conditions under which four columns have the product estimates we need. Suppose that orthogonal positive contractions $e, f \in C$ and invertibles $Z, W \in \tilde{C}$ satisfy

$$ge = e, \quad gf = f, \quad Z^*W = 1, \quad Z = 1 + gzg, \quad W = 1 + gwg, \quad \|Z\|, \|W\| \leq 2 \quad (9.4)$$

for some $z, w \in C$. For such a choice, define

$$\begin{aligned} X_0 &= aL(u_*), \quad X = \ell^{-1/2}avRZe, \\ Y_0 &= bL(u), \quad Y = \ell^{-1/2}bRWf. \end{aligned} \quad (9.5)$$

These elements belong to C : the only polar factor is $vR \in C$, and all unitized factors occur multiplied by elements of C . For any such admissible choice, the proposed outputs are

$$Q_a = (X_0X_0^* + XX^*)^{1/2}, \quad Q_b = (Y_0Y_0^* + YY^*)^{1/2}. \quad (9.6)$$

The norm of each positive output dominates the norms of its two columns in every primitive quotient. At the end, a row polar decomposition will bound the product of these outputs by the four cross products estimated next.

Put $E_R = RuR - \ell$. Since $E_Rg = 0$ and $f = gf$, we have

$$E_RWf = E_Rf + E_Rgwgf = 0.$$

In particular, no commutation of W with g is needed. Using $v^*ab = u$ and retaining the indicated order of factors gives

$$X^*Y = \ell^{-1}eZ^*RuRWf = eZ^*Wf = ef = 0. \quad (9.7)$$

The other three products obey

$$\|X_0^*Y_0\| \leq 6\ell, \quad \|X^*Y_0\| \leq 2\sqrt{6\ell}, \quad \|X_0^*Y\| \leq 2\sqrt{6\ell}. \quad (9.8)$$

Indeed,

$$\begin{aligned} X_0^*Y_0 &= vL(u)uL(u), \\ X^*Y_0 &= \ell^{-1/2}eZ^*RuL(u), \\ X_0^*Y &= \ell^{-1/2}vL(u)uRWf. \end{aligned}$$

Here $\|uL(u)\| \leq 6\ell$. For $0 \leq s \leq \ell$, $\ell^{-1/2}R(s)sL(s) \leq \sqrt{\ell}$; for $s \geq \ell$ the same expression equals $\sqrt{s}L(s) \leq \sqrt{6\ell}$. Together with (9.4), these scalar estimates prove (9.8). We now construct e, f, Z, W so that, in each prescribed primitive quotient, at least one of the two relevant columns also has a fixed lower norm bound.

Two orthogonal cores. Set $h_0 = (u - 2\ell)_+$. Since $h_0 \oplus h_0 \precsim h_0$, Lemma 7.6 supplies a row $[r_1, r_2] \in M_{1,2}(C)$ such that

$$r_i^* r_j = \begin{cases} \theta(h_0), & i = j, \\ 0, & i \neq j, \end{cases} \quad r_1 r_1^* + r_2 r_2^* \in \text{Her}(h_0), \quad (9.9)$$

where $\theta: [0, \infty) \rightarrow [0, 1]$ is continuous, vanishes near zero, and equals one on $[2\ell, \infty)$. Define

$$e_0 = r_1 r_1^*, \quad f_0 = r_2 r_2^*.$$

They are orthogonal positive contractions in $\text{Her}(h_0)$; each belongs to this hereditary algebra because it is dominated by their sum. In any primitive quotient with $\|u_P\| \geq 4\ell$, $\|(h_0)_P\| \geq 2\ell$, and hence

$$\|(e_0)_P\| = \|(f_0)_P\| = \|\theta((h_0)_P)\| = 1. \quad (9.10)$$

This includes equality at the threshold 4ℓ by the chosen plateau.

Put

$$k_e = (e_0 - 1/2)_+, \quad k_f = (f_0 - 1/2)_+,$$

and let $e = \chi(e_0)$ and $f = \chi(f_0)$, where $\chi: [0, 1] \rightarrow [0, 1]$ is continuous, vanishes near zero, and is one on $[1/2, 1]$. Then

$$ef = 0, \quad ek_e = k_e, \quad fk_f = k_f, \quad ge = e, \quad gf = f. \quad (9.11)$$

For the last two identities, $gh_0 = h_0$ implies that multiplication by g is the identity on $\text{Her}(h_0)$. By (9.10), both $(k_e)_P$ and $(k_f)_P$ are nonzero whenever $\|u_P\| \geq 4\ell$.

The quotient regions requiring a high column. Define

$$\tau = \frac{r\sqrt{\ell}}{8}, \quad C_1 = avR, \quad C_2 = bR, \quad h_i = (|C_i g| - \tau)_+, \quad d_i = (|C_i N| - 2\tau)_+. \quad (9.12)$$

We first check why a lower bound on the region $\|(d_i)_P\| \geq \tau$ will recover any signal missed by the corresponding low column. Suppose $\|a_P\| \geq r$ and $\|(X_0)_P\| < r/2$. Since $a = aL(u_*) + aN(u_*)$, we have $\|(aN(u_*))_P\| > r/2$. The identity $N(u_*) = vNv^*$ gives $\|(avN)_P\| > r/2$. On $[0, 1]$, the scalar inequality $R^2 N^2 \geq \ell N^2$ holds: where N is nonzero, $s > 5\ell$ and $R(s)^2 = \ell/s \geq \ell$. Consequently

$$(C_1 N)(C_1 N)^* = avR^2 N^2 v^* a \geq \ell avN^2 v^* a,$$

and therefore

$$\|(C_1 N)_P\| > \frac{r\sqrt{\ell}}{2} = 4\tau.$$

It follows that $\|(d_1)_P\| > 2\tau$. Similarly, if $\|b_P\| \geq r$ and $\|(Y_0)_P\| < r/2$, then $\|(bN)_P\| > r/2$. The order inequality $bR^2 N^2 b \geq \ell bN^2 b$ gives $\|(d_2)_P\| > 2\tau$. Thus every prescribed quotient either has a low column of norm at least $r/2$ or belongs to the corresponding perturbation region.

Comparison inputs and normalized left factors. We now verify the inputs to Proposition 7.8 on these two regions. If $(d_i)_P \neq 0$, then $C_i N = C_i gN$ implies $\|(C_i g)_P\| > 2\tau$, so $(h_i)_P \neq 0$. The same assumption implies $N_P \neq 0$, and therefore $\|u_P\| > 5\ell$. Consequently $(k_e)_P, (k_f)_P \neq 0$. The primitive ideal test and Corollary 8.6 give

$$d_1 \precsim h_1, \quad d_1 \precsim k_e, \quad d_2 \precsim h_2, \quad d_2 \precsim k_f. \quad (9.13)$$

Let $D_i = C_i g$, and use continuous functional calculus in the unitization to set

$$F_i = (\max\{\tau, |D_i^*|\})^{-1}, \quad p_i = F_i D_i. \quad (9.14)$$

Thus $\|F_i\| \leq \tau^{-1}$ and $p_i \in C$. The functional-calculus intertwining identity

$$F_i D_i = D_i (\max\{\tau, |D_i|\})^{-1}$$

follows first for polynomials in $D_i D_i^*$ and $D_i^* D_i$, and then by uniform approximation. In particular,

$$p_i^* p_i = |D_i|^2 (\max\{\tau, |D_i|\})^{-2} \leq 1, \quad p_i^* p_i h_i = h_i. \quad (9.15)$$

The use of $|D_i^*|$ in the left factor F_i is essential to this identity.

Two compatible inverse perturbations. Fix

$$\eta = \frac{1}{8}, \quad \eta' = \frac{\sigma\eta}{16} = \frac{\sigma}{128}.$$

Apply Proposition 7.8, with $n = 1$, to

$$(p, q, h, k, c, T, \alpha) = (p_1, e, h_1, k_e, d_1, F_1 C_1 e, \tau)$$

at scale η . Its hypotheses follow from (9.11), (9.13), and (9.15). For the resulting contraction $y \in C$, put

$$Z_1 = 1 + \eta g y g.$$

Since $\|Z_1 - 1\| \leq \eta < 1$, the element Z_1 is invertible; define $W_1 = (Z_1^*)^{-1}$. Since $g e = e$, the conclusion of the Proposition becomes

$$\|(F_1 C_1 Z_1 e)_P\| \geq \sigma\eta \quad \text{if } \|(d_1)_P\| \geq \tau. \quad (9.16)$$

Next apply the same Proposition at scale η' to

$$(p, q, h, k, c, T, \alpha) = (p_2, f, h_2, k_f, d_2, F_2 C_2 W_1 f, \tau).$$

The element T is in C ; the Proposition imposes no bound on its norm. With the resulting contraction $y' \in C$, set

$$W = W_1 + \eta' g y' g.$$

Since $g f = f$, the second conclusion is

$$\|(F_2 C_2 W f)_P\| \geq \sigma\eta' \quad \text{if } \|(d_2)_P\| \geq \tau. \quad (9.17)$$

We record the inverse estimates, including the factorizations needed to preserve (9.16). For $a_0 \in C$ with $\|a_0\| < 1$, the Neumann series gives

$$(1 + g a_0 g)^{-1} = 1 + g \left(\sum_{k=1}^{\infty} (-1)^k a_0 (g^2 a_0)^{k-1} \right) g. \quad (9.18)$$

The series of middle factors converges in C with norm at most $\|a_0\|/(1 - \|a_0\|)$. This formula does not require g to be a projection. Applying it to Z_1^* gives

$$W_1 = 1 + g w_1 g, \quad \|w_1\| \leq \frac{1}{7}.$$

Thus $W = 1 + gw g$ for $w = w_1 + \eta' y' \in C$, with

$$\|w\| \leq \frac{1}{7} + \frac{1}{128} < \frac{1}{4}.$$

Applying (9.18) to $W^* = 1 + gw^* g$ now proves that W is invertible. Define $Z = (W^*)^{-1}$. The same series gives

$$Z = 1 + gzg, \quad \|z\| \leq \frac{1}{3}, \quad \|Z\| \leq \frac{4}{3}, \quad \|W\| \leq \frac{5}{4}, \quad \|Z_1\| \leq \frac{9}{8}. \quad (9.19)$$

In particular, all conditions in (9.4) hold.

The second perturbation changes the inverse used in the first column. To keep the first lower bound independent of ℓ , we must estimate this change without the potentially large factor $\|F_1\|$. The inverse difference has the exact order

$$Z - Z_1 = -\eta' Zg(y')^* gZ_1.$$

As $Zg = g(1 + zg^2)$, its effect on the first normalized column is bounded by

$$\begin{aligned} \|F_1 C_1(Z - Z_1)e\| &= \eta' \|p_1(1 + zg^2)(y')^* gZ_1 e\| \\ &\leq \eta' \left(1 + \frac{1}{3}\right) \frac{9}{8} = \frac{3}{2} \eta'. \end{aligned} \quad (9.20)$$

Since $3\eta'/2 \leq \sigma\eta/2$, (9.16) remains bounded below by $\sigma\eta/2$ with Z in place of Z_1 . Using $\|F_i\| \leq \tau^{-1}$ in each primitive quotient, and $\tau\ell^{-1/2} = r/8$, we obtain

$$\begin{aligned} \|X_P\| &\geq \frac{r}{8} \frac{\sigma\eta}{2} && \text{if } \|(d_1)_P\| \geq \tau, \\ \|Y_P\| &\geq \frac{r}{8} \sigma\eta' && \text{if } \|(d_2)_P\| \geq \tau. \end{aligned} \quad (9.21)$$

The cancellation $\tau\ell^{-1/2} = r/8$ is what removes ℓ from both lower bounds.

Assembling the two positive outputs.

To check the hereditary membership of Q_a, Q_b , (9.5) and the bounds on Z, W give

$$X_0 X_0^* \leq a^2, \quad X X^* \leq 4\ell^{-1} a^2, \quad Y_0 Y_0^* \leq b^2, \quad Y Y^* \leq 4\ell^{-1} b^2.$$

By Lemma 2.1, the first two summands belong to $\text{Her}(a)$ and the last two to $\text{Her}(b)$. Thus $Q_a \in \text{Her}(a)$ and $Q_b \in \text{Her}(b)$. The earlier coverage reduction and (9.21), together with domination of the column norms by the output norms, prove (9.1) with

$$m = \frac{r}{8} \sigma\eta' = \frac{r\sigma^2}{1024}.$$

Here $m \leq r/2$ and $\eta' \leq \eta/2$, so this single lower bound applies to all cases. It is independent of ℓ .

Finally put $U_a = [X_0, X]$ and $U_b = [Y_0, Y]$ in $M_{1,2}(C)$. Their range squares are Q_a^2 and Q_b^2 . The rectangular polar decompositions in the bidual have the form $U_a = Q_a v_a$ and $U_b = Q_b v_b$, where $v_a v_a^*$ and $v_b v_b^*$ are the support projections of Q_a and Q_b , respectively. Hence both identities

$$U_a^* U_b = v_a^* Q_a Q_b v_b, \quad Q_a Q_b = v_a (U_a^* U_b) v_b^*$$

hold. Since partial isometries have norm at most one, they imply

$$\|Q_a Q_b\| = \|U_a^* U_b\| \leq 6\ell + 4\sqrt{6\ell} \leq 20\sqrt{\ell},$$

by (9.7) and (9.8). This proves (9.2).

The construction also covers zero elements and empty norm regions. If $h_0 = 0$, the row in (9.9) may be zero. Empty perturbation targets permit zero perturbations, and the coverage argument then assigns every prescribed quotient to a low column. All inverse operations have either the positive floor τ or the convergent series (9.18). $\square$

The norms of the preceding outputs may grow as ℓ decreases. The next observation controls the comparison coefficients using only the fixed spectral floor m . This is enough for their small product to survive Cuntz implementation.

Lemma 9.2 (Comparison above a spectral floor). *Let C be a C^* -algebra, let $Q, c \in C_+$ with $\|c\| \leq 1$, and let $m > 0$. If $c \precsim (Q^2 - m^2/4)_+$, then for each $0 < \delta < 1$ there is $x \in C$ such that*

$$\|(Qx)^*(Qx) - c\| < \delta, \quad \|Qx\| \leq 2, \quad \|x\| \leq \frac{4}{m}. \quad (9.22)$$

Proof. Choose a comparison implementer $v \in C$ with

$$\|v^*(Q^2 - m^2/4)_+v - c\| < \delta.$$

Such an implementer lies in C by compression to the two scalar matrix corners. On $[0, \|Q\|]$ define the continuous function

$$j(s) = \begin{cases} s^{-1}(s^2 - m^2/4)_+^{1/2}, & s > 0, \\ 0, & s = 0, \end{cases} \quad x = j(Q)v.$$

Then $(Qx)^*(Qx) = v^*(Q^2 - m^2/4)_+v$ and $\|Qx\|^2 < 1 + \delta < 2$, which in particular gives $\|Qx\| \leq 2$. Moreover the scalar inequality

$$j(s)^2 \leq \frac{4}{m^2} s^2 j(s)^2$$

holds because j vanishes on $[0, m/2]$. Functional calculus followed by multiplication by v^* and v gives

$$x^*x \leq \frac{4}{m^2} (Qx)^*(Qx).$$

Thus $\|x\| \leq (2/m)\|Qx\| \leq 4/m$, independently of the norm of the original comparison implementer v . $\square$

Proposition 9.3 (Direct exact proof). *Let C be an exact C^* -algebra satisfying (P). For every $a, b \in C_+$ and every $\varepsilon > 0$, there are $s, t \in C$ such that*

$$\|s^*a^2s - a^2\| < \varepsilon, \quad \|t^*b^2t - b^2\| < \varepsilon, \quad \|s^*abt\| < \varepsilon.$$

Consequently C is strongly purely infinite.

Proof. Fix the constant σ in Proposition 7.8 with $n = 1$. Let $a, b \in C_+$ be contractions and let $\varepsilon > 0$. Choose $0 < r < 1$ with $r^2 < \varepsilon/3$ and put $m = r\sigma^2/1024$. Next choose $0 < \ell < 1/10$ so small that

$$\frac{320\sqrt{\ell}}{m^2} < \frac{\varepsilon}{3}. \quad (9.23)$$

Take Q_a, Q_b from Lemma 9.1. If $(a^2 - r^2)_+$ has nonzero image at P , then $\|a_P\| > r$, so $\|(Q_a)_P\| \geq m$. Thus $(Q_a^2 - m^2/4)_+$ has nonzero image there. The primitive ideal test and Corollary 8.6 imply

$$(a^2 - r^2)_+ \precsim (Q_a^2 - m^2/4)_+, \quad (b^2 - r^2)_+ \precsim (Q_b^2 - m^2/4)_+. \quad (9.24)$$

Finally choose $0 < \delta < 1$ such that

$$\delta + (4 + \delta)\delta < \frac{\varepsilon}{3}. \quad (9.25)$$

Apply Lemma 9.2 to the comparisons in (9.24). We obtain $x_a, x_b \in C$ with

$$\begin{aligned} \|(Q_a x_a)^*(Q_a x_a) - (a^2 - r^2)_+\| &< \delta, \\ \|(Q_b x_b)^*(Q_b x_b) - (b^2 - r^2)_+\| &< \delta, \end{aligned}$$

and

$$\|Q_a x_a\|, \|Q_b x_b\| \leq 2, \quad \|x_a\|, \|x_b\| \leq \frac{4}{m}.$$

Consequently

$$\|x_a^* Q_a Q_b x_b\| \leq \frac{16}{m^2} \|Q_a Q_b\| \leq \frac{320\sqrt{\ell}}{m^2}. \quad (9.26)$$

Since $Q_a \in \text{Her}(a) = \overline{aCa}$, $Q_a x_a \in \overline{aC}$; similarly $Q_b x_b \in \overline{bC}$. Choose $s, t \in C$ such that

$$\|as - Q_a x_a\| < \delta, \quad \|bt - Q_b x_b\| < \delta.$$

For elements U, V of norm at most 2, replacing them by elements within δ changes U^*U or U^*V by at most $(4 + \delta)\delta$, by expanding the difference and using the triangle inequality. Also $\|a^2 - (a^2 - r^2)_+\|, \|b^2 - (b^2 - r^2)_+\| \leq r^2$. Therefore

$$\begin{aligned} \|s^* a^2 s - a^2\|, \|t^* b^2 t - b^2\| &\leq r^2 + \delta + (4 + \delta)\delta < \varepsilon, \\ \|s^* a b t\| &\leq \frac{320\sqrt{\ell}}{m^2} + (4 + \delta)\delta < \varepsilon, \end{aligned}$$

by (9.23) and (9.25). The choice order is r , then m , then ℓ , then δ ; the lower bound m has no dependence on the later parameters.

For arbitrary positives a, b , put $M = \max\{1, \|a\|, \|b\|\}$. Apply the preceding result to $a/M, b/M$ with tolerance ε/M^2 . Multiplication of all three inequalities by M^2 gives the required inequalities for a, b with the same $s, t \in C$. Zero elements, including the zero algebra, are included. These are precisely the three inequalities in (1.1), so C is strongly purely infinite. $\square$

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